

The Fine-Structure Constant from the Boundary of Observability: A Rank-5 Non-Invertible Projection

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Abstract

The fine-structure constant $\alpha^{-1} \approx 137.036$ enters the Standard Model as a measured input. This paper derives its value from a single declared import: a Majorana conservation law read under restricted observational access. The construction is a non-invertible projection from the five-dimensional domain $\mathbb{R}^4 \times S^1(\tau)$ onto an observable four-dimensional boundary; the projection rank is not chosen but fixed at $r = 5$ by a two-sided exclusion theorem (conservation, autosimilar closure, metric stability). The resulting boundary filter has the exact static pole $\alpha_{\text{QGT}}^{-1} = 70 + 30\sqrt{5}$; two structural corrections, both parameter-free, shift this to $\alpha^{-1} = 137.035999170$, within 0.3σ of CODATA 2022. This is the value at the QED/atomic threshold: the boundary input to the perturbative renormalisation group, not a result of it. The same structural objects give $\sin^2 \theta_W = 0.23125$, within 0.75σ of the Z -pole value; electroweak running and scheme conversion remain outside the present derivation. Every step is closed-form arithmetic, verified by an accompanying script that includes the counterfactual ranks.

Keywords: fine-structure constant; non-invertible projection; Majorana spinor; rank-5 boundary filter; Hilbert quadrature; fundamental constants; gauge structure.

1 Introduction

The dimensionless constant $\alpha^{-1} \approx 137.036$ introduced by Sommerfeld[1] governs every electromagnetic interaction. Its value is not determined by the Standard Model, which treats it as a measured input. That this should be so is, on reflection, not a gap in our calculations but a structural fact about the framework itself.

This paper studies the fine-structure constant as the stable operating point of a rank-5 boundary filter induced by a non-invertible projection. The construction starts from the Noether current $J^I = \bar{\Xi} \Gamma^I \Xi$ of a Majorana spinor field Ξ under restricted observational access on a five-dimensional domain $\mathbb{R}^4 \times S^1(\tau)$ and derives the projection rank, the boundary filter, and the dynamical corrections that connect the static projection pole to the observed value of α^{-1} .

1.1 Structure of the argument

The paper proceeds in three steps.

First, it shows that no admissibility projection with non-trivial physical kernel can be derived from within an $\mathbb{R}^4$ -based field theory (Section 2). This is not a limitation of any specific model: it is a structural fact about any framework that cannot identify, from within, which of its own degrees of freedom are observable in principle.

Second, it constructs the minimal non-invertible projection that closes this gap (Section 3). Three requirements (a single conserved flow, no pre-existing metric, a non-trivial boundary algebra) select uniquely the five-dimensional domain $\mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau)$, the Majorana bulk field $\Xi = C\bar{\Xi}$, and the projection $\Pi = \int_{S^1} d\tau$. The Majorana self-conjugacy condition $\Xi = C\bar{\Xi}$ eliminates all cross-terms in

the bilinear norm of the mode occupancies, leaving a second-order recursion $|O_{n+2}|^2 = a|O_{n+1}|^2 + b|O_n|^2$. Its characteristic polynomial must be $\lambda^2 - \lambda - 1 = 0$, the autosimilar equation $\varphi^2 = \varphi + 1$ whose roots φ and $-1/\varphi$ are the eigenvalues of the scale operator on $S^1(\tau)$. This fixes $a = b = 1$ uniquely: the Fibonacci recursion is not a choice but the only transport law whose characteristic polynomial is palindromic and compatible with the autosimilar scale φ of the fibre. The rank $r = 5$ follows from the unique non-trivial fixed point of the Fibonacci sequence: $F_{n^*} = n^*$ has exactly one solution with prime holonomy, $n^* = 5$.

Third, given this minimal structure, the conservation law $\partial_I J^I = 0$ decomposes on the boundary into two chiral channels (ξ, η) whose stable product $\xi\eta = \alpha^{-1}$ is fixed by the rank-5 boundary filter. The static projection pole is

$$\alpha_{\text{QGT}}^{-1} = (F_5 + F_4\sqrt{5})^2 = (5 + 3\sqrt{5})^2 = 70 + 30\sqrt{5},$$

where $F_4 = 3$ and $F_5 = 5$ are the Fibonacci numbers at the fixed point. Two structural corrections shift the static pole to the observed value (Section 5): the holonomy residue of the unique confined mode $n = 7$, and the breathing back-action of the projected boundary, both derived without free parameters. The result lies within 7×10^{-9} of the CODATA 2022 value, well within its experimental uncertainty. The closed results of the paper are the α -sector and the one-step structural electroweak boundary matching of Section 6. Fermionic transport beyond the charged-lepton sector, QCD dynamics, the full scheme-dependent electroweak renormalisation-group evolution, and the gravitational sector are treated only as open extensions.

The continuous bulk (a conserved phase on S^1 , an autosimilar scale φ , a Majorana spinor) generates under restricted projection something irreducibly discrete: five modes, the catalogue $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$, the integer matrix $\mathbf{M}$, the value $\alpha^{-1} \in \mathbb{Z}[\sqrt{5}]$. The discrete structure does not enter from outside. It is the residue of a continuous coherence that cannot cross the projection interface completely. What that coherence could not carry through left, as its residue, $\alpha^{-1} = 137.036$.

2 The gap: observability and its limits in $\mathbb{R}^4$

Sommerfeld introduced α in 1916 as the ratio of the electron velocity in the first Bohr orbit to the speed of light. One hundred and ten years later, its value remains a measured input in every field-theoretic framework built on $\mathbb{R}^4$: the Standard Model uses it but does not derive it. This section shows why this is structurally inevitable, and what kind of structure would be needed to change it.

2.1 The gap in every framework built on $\mathbb{R}^4$

Modern field theories specify how physical quantities evolve once the relevant degrees of freedom are identified, but they do not provide an intrinsic criterion for determining which quantities are observable in principle. Observables are simply those that are formally definable and are assumed accessible given sufficient experimental refinement. This assumption is never examined at a structural level.

What would it mean for such a criterion to exist? It would mean that there is a map $E : \mathcal{A} \rightarrow \mathcal{A}_{\text{adm}}$ from the full observable algebra to an admissible subalgebra, satisfying four conditions simultaneously: $\mathcal{A}_{\text{adm}}$ is a subalgebra of $\mathcal{A}$; E is idempotent ($E \circ E = E$); E has a non-trivial kernel ($\ker E \neq \{0\}$); and the kernel consists of degrees of freedom physically eliminated prior to observation, not merely gauge-redundant. Restricted observational access means precisely this: the boundary observer cannot reconstruct bulk invariants from boundary data alone, because the map from bulk to boundary is not invertible. The last condition distinguishes a genuine admissibility criterion from a gauge fixing: a gauge fixing removes descriptive redundancy while preserving all physical content.

For such a map to have a non-trivial kernel, it must be non-invertible. This is not optional. If E were invertible, every element of $\mathcal{A}$ could be reconstructed from $\mathcal{A}_{\text{adm}}$ via E^{-1} : in the idealised limit of complete measurement, supposedly inadmissible degrees of freedom would become accessible, contradicting their inadmissibility. A genuine admissibility criterion must therefore be non-invertible.

2.2 No such criterion exists within SM/GR-like theories

We now show that no mechanism available within Standard-Model and General-Relativity-like frameworks satisfies these four conditions simultaneously.

Gauge fixing selects one representative per gauge orbit, reducing $\mathcal{A}$ to gauge-invariant quantities. Its kernel contains directions of descriptive redundancy, not physical degrees of freedom: the last condition fails.

Coarse-graining replaces fine-grained degrees of freedom with effective ones. It introduces practical inaccessibility, not principled inadmissibility: in the idealised limit of complete experimental control, the eliminated degrees of freedom become accessible again. The map is not a genuine admissibility criterion.

Decoherence suppresses quantum coherences under environmental coupling. It is a dynamical process depending on the specific environment and coupling: it is not a fixed intrinsic map on $\mathcal{A}$.

Renormalisation re-expresses the theory at a different scale without identifying degrees of freedom that are inadmissible across all scales.

Algebraic quantum field theory specifies observable algebras through additional axioms — locality, Haag duality, and related conditions. These are physically motivated but externally imposed: they are not derived from the field-theoretic dynamics and therefore do not constitute an intrinsic admissibility criterion.

Since no standard internal mechanism considered above satisfies all four conditions simultaneously, no admissibility projection can be derived from within SM/GR-like frameworks: the required structure is absent from any theory formulated purely on $\mathbb{R}^4$.

2.3 The consequence for α

A dimensionless constant derivable as a structural invariant of $\mathcal{A}$ would be a ratio of admissible quantities determined by the algebra's internal structure. But if the admissible subalgebra is not fixed by any intrinsic criterion, no internal criterion fixes which ratios are physically meaningful. Therefore no dimensionless coupling constant (in particular α) can be derived from within SM/GR-like frameworks. It enters as an external parameter because the structure required to make it a derived quantity does not exist within $\mathbb{R}^4$.

This result has a sharper algebraic form. Every $*$ -automorphism $\phi : \mathcal{A} \rightarrow \mathcal{A}$ of a C^* -algebra is isometric: the C^* -identity $\|a^*a\| = \|a\|^2$ is preserved under ϕ , giving $\|\phi(a)\| = \|a\|$ for all a , hence $\ker \phi = \{0\}$. [2] An intrinsic admissibility criterion realised as an automorphism would therefore have trivial kernel, contradicting the requirement $\ker E \neq \{0\}$. This confirms, in the algebraic category, the structural conclusion above.

2.4 What follows from the no-go

The argument above is not a philosophical remark: it closes a physical question. If any dimensionless constant of nature is to have a structural origin rather than entering as a measured input, the theory must contain a non-invertible map from its full algebra to an admissible subalgebra with a non-trivial physical kernel. No framework built intrinsically on $\mathbb{R}^4$ contains such a map.

What must exist, therefore, is a structure that supplies it from outside $\mathbb{R}^4$, a projection Π from a larger domain, irreversible and non-invertible, whose kernel carries the physically eliminated degrees of freedom. The construction developed in the present paper begins here: given such a projection, what is the minimal construction consistent with a single conserved flow, no pre-existing metric, and a non-trivial boundary algebra? Section 3 answers this. The value of α^{-1} then follows from the finite-rank structure of that minimal projection, not as an input, but as the unique stable operating point of its boundary filter.

The minimal construction satisfying these three requirements is the subject of Section 3:

- (i) a single globally conserved flow (one Noether current, not two);

- (ii) no pre-existing metric (the observable metric of $\mathbb{R}^4$ must be a derived output, not a structural input);
 - (iii) a non-trivial boundary algebra (the projection must produce observables, not merely a quotient).
- Within this construction, α^{-1} is not a parameter: it is the unique stable operating point of the boundary filter forced by the finite-rank structure of the minimal projection.

3 The minimal construction

3.1 The zero mode: the topological carrier of a conserved flow

Requirement (i) is a single globally conserved flow. A conserved current J^I must be self-consistent: if it is transported around any closed path in the domain, it must recognise itself. For a global conservation law $\partial_I J^I = 0$ to be self-consistent under restricted observational access, the current must carry a compact internal phase, a *zero mode* $\chi \in S^1$, that closes coherently under transport.

This is the minimal axiom consistent with a globally conserved flow under restricted accessibility: every subsequent step follows from it together with requirements (i)–(iii).

3.2 The bulk manifold: selected by the minimality assumptions

The zero mode $\chi \in S^1$ requires a domain that houses a compact internal phase together with four extended directions of observable bilinears. The four extended directions are required by requirement (iii): fewer would not support a non-trivial boundary algebra; a compact extended domain would enforce global identifications inconsistent with (iii).

Additional compact dimensions beyond S^1 would introduce independent conserved channels, violating requirement (i). Non-trivial fibrations would couple the kernel to the boundary in ways that destroy the separation between admissible and inadmissible observables, violating (ii).

By exhaustion of alternatives *within the three minimality requirements (i)–(iii)*, the minimal domain is:

$$\mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau). \quad (1)$$

The coordinate τ parametrises transport around the compact fibre. It is not the physical time of $\mathbb{R}^4$: physical time emerges only at the boundary, as the irreversible ordering of the phase between the two chiral channels ξ and η after projection.

The structure $\mathbb{R}^4 \times S^1$ appears in Kaluza–Klein theory, but the content is different: in Kaluza–Klein, the bulk metric is the input and gauge fields emerge as outputs. Here requirement (ii) excludes any pre-existing metric. The projection Π is non-invertible, and the observable metric of $\mathbb{R}^4$ is a derived stability condition of the lossy boundary.

3.3 The bulk field: selected by the minimality assumptions

What field lives on $\mathcal{M}^5$? Requirement (ii) excludes any field that requires a pre-existing metric to define its dynamics. The field must also carry the zero mode coherently around $S^1(\tau)$, carry no net bulk charge (since a single conserved flow, requirement (i), admits no independent charge sector), and have non-trivial holonomy so that the projection Π produces a non-trivial observable algebra (requirement (iii)).

A single-valued carrier ($\hat{H} = +1$ under 2π transport) has trivial holonomy and produces no observable structure under Π : excluded by (iii). Vector fields require a pre-existing metric: excluded by (ii). Dirac spinors carry net charge ($\psi \neq C\bar{\psi}$): excluded by (i). Weyl spinors are globally incomplete on $S^1(\tau)$: the two Weyl sectors are independent and the self-referential closure required by a single conserved flow is absent.

Within these assumptions, the minimal admissible field is the Majorana spinor:

$$\Xi = C\bar{\Xi}, \quad \hat{H} = -1 \text{ (} 2\pi\text{-holonomy)}. \quad (2)$$

Real, globally neutral, double-valued: the exact self-conjugacy that a single conserved flow under restricted access demands.

The minimal construction is therefore:

$$\mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau), \quad \Xi = C\bar{\Xi} \text{ (Majorana)}, \quad \Pi = \int_{S^1} d\tau. \quad (3)$$

Within this construction, all remaining structure is a consequence.

3.4 The autosimilar scale: the unique self-similar ratio

The compact fibre $S^1(\tau)$ supports a hierarchy of modes. For the hierarchy to be self-consistent under the projection, each level structurally identical to every other under rescaling, the ratio x between adjacent modes must satisfy $x^2 = x + 1$. This has the unique positive solution:

$$\varphi = \frac{1 + \sqrt{5}}{2}. \quad (4)$$

φ is not a free parameter. It is the unique fixed point of the map $x \mapsto 1 + 1/x$ on the positive reals: the ratio that any self-similar hierarchy must carry.

3.5 The Clifford extension and spectral curvature

Transporting Ξ along $S^1(\tau)$ requires extending the Clifford algebra $\text{Cl}(1, 3)$ to $\text{Cl}(4, 1)$. The minimal extension introduces the canonical involution Γ_6 satisfying $\Gamma_6^2 = 1$ and $\{\Gamma_6, \gamma^\mu\} = 0$. The chiral projectors onto the two Majorana sectors are:

$$P_{\varphi\pm} = \frac{1}{2}(1 \pm \Gamma_6), \quad (5)$$

with $P_{\varphi+} + P_{\varphi-} = 1$ and $P_{\varphi+}P_{\varphi-} = 0$.

For the two chiral branches to close coherently on $S^1(\tau)$, the holonomy amplitude of the projection must equal the autosimilar scale φ . This forces the spectral curvature:

$$\kappa = \sqrt{5} = \varphi + \frac{1}{\varphi}. \quad (6)$$

The spectral generator of the projection is the rescaled involution:

$$\gamma^6 := \kappa \Gamma_6 = \sqrt{5} \Gamma_6, \quad (\gamma^6)^2 = 5, \quad (7)$$

so that $P_{\varphi\pm} = \frac{1}{2}(1 \pm \gamma^6/\sqrt{5})$. The temporal connection follows: $\Gamma_\tau = \varphi/\kappa = \varphi/\sqrt{5}$.

3.6 Fibonacci structure: consequence, not input

The Majorana condition $\Xi = C\bar{\Xi}$ eliminates all cross-terms in the norm of the temporal mode occupancies $O_n = \langle \Xi, v_n \rangle$. The surviving diagonal terms satisfy the recursion:

$$|O_{n+2}|^2 = |O_{n+1}|^2 + |O_n|^2. \quad (8)$$

This is not imposed: it is the unique second-order closure recursion whose characteristic polynomial $\lambda^2 - \lambda - 1 = 0$ has roots φ and $-1/\varphi$, the eigenvalues of γ^6 in scale space.

Proposition 1 (Majorana forces $F_r = r$, unique solution $r = 5$). *The Majorana self-conjugacy eliminates all cross-terms in the bilinear norm of the temporal mode occupancies, leaving a second-order recursion $|O_{n+2}|^2 = a|O_{n+1}|^2 + b|O_n|^2$. The minimal scale-free closure (no preferred scale introduced by the coefficients) forces $a = b$; Majorana symmetry of the two chiral branches forces $a = b$; and normalisation to unit holonomy at the autosimilar scale φ forces $a = b = 1$. Therefore $|O_n|^2 = F_n$ (the n -th Fibonacci number). For the projection rank r to equal its own holonomy norm, one requires $F_r = r$. Among all positive integers: $F_1 = 1 = 1$ (trivial), $F_5 = 5 = 5$ (unique non-trivial), and $F_r > r$ for all $r > 5$. Therefore $r = 5$ is the unique non-trivial rank at which the Majorana holonomy self-consistently closes the projection.*

Proof sketch. The Fibonacci sequence grows as $F_n \sim \varphi^n / \sqrt{5}$. For $n > 5$: $F_6 = 8 > 6$, $F_7 = 13 > 7$, and by induction $F_n > n$ for all $n \geq 6$ (since $F_{n+1} = F_n + F_{n-1} > n + (n-1) = 2n-1 > n+1$ for $n \geq 3$). For $n < 5$: $F_2 = 1 < 2$, $F_3 = 2 < 3$, $F_4 = 3 < 4$. The only solutions of $F_r = r$ are $r = 1$ (trivial, zero-dimensional boundary) and $r = 5$ (non-trivial). Primality of $r = 5$ further rigidifies the selection: 5 is the unique prime satisfying $F_p = p$. $\square$

Fibonacci structure is therefore a theorem, not an input. It is the spectral algebra of the Majorana chiral projector on $S^1(\tau)$.

Proposition 2 (Uniqueness of the minimal construction). *Under the three requirements (i)–(iii), the minimal construction $(\mathcal{M}^5, \Xi, \Pi)$ is unique up to diffeomorphism of the compact fibre.*

Proof. Manifold. Requirement (i) forces exactly one compact direction (additional compact directions introduce independent conserved channels). Requirement (iii) forces at least four extended directions (fewer cannot support a non-trivial observable algebra). Non-trivial fibrations violate (ii) by coupling kernel to boundary. Hence $\mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau)$ is the unique minimal manifold.

Field. Single-valued carriers have trivial holonomy (excluded by (iii)). Vector fields require a pre-existing metric (excluded by (ii)). Dirac spinors carry net charge (excluded by (i)). Weyl spinors are globally incomplete on $S^1(\tau)$ and cannot close the single-flow self-reference. The Majorana spinor $\Xi = C\bar{\Xi}$ is the unique remaining admissible field.

Projection. The rank is uniquely $r = 5$ by Proposition 1. The autosimilar scale is uniquely φ (unique positive solution of $x^2 = x + 1$). The spectral curvature is uniquely $\kappa = \sqrt{5} = \varphi + 1/\varphi$. No degree of freedom remains. $\square$

3.7 The kernel and what it means

The conservation law $\partial_I J^I = 0$ imposed on this structure decomposes as $\partial_\tau J^\tau + \partial_\mu J^\mu = 0$. Restricted accessibility requires $\ker \Pi \neq \{0\}$: not all invariants of J^I are operationally available to the boundary observer, as established in §2. The zero mode cannot cross the projection interface completely. What does not cross returns as non-linear boundary tension.

Within the minimal construction (3):

$S^1(\tau) \rightarrow \partial_I J^I = 0 \rightarrow \Xi = C\bar{\Xi}, (\gamma^6)^2 = 5, \kappa = \sqrt{5}, \Gamma_\tau = \varphi/\sqrt{5} \rightarrow \ker \Pi \neq \{0\} \rightarrow$ boundary physics.

Each step follows from the one before within the construction. None is assumed independently.

Remark 1 (Kernel, lossiness, and the problem of time). The boundary $\text{Im } \Pi$ is lossy: it cannot be inverted to reconstruct the bulk. The non-invertibility of Π is the projection-theoretic signature of bulk-boundary entanglement: $\ker \Pi$ contains precisely the degrees of freedom that are entangled with the boundary but cannot be reconstructed from boundary data alone. The constants of $\mathbb{R}^4$ are stability conditions of this lossy boundary, not free parameters. In the bulk, $\mathcal{D}_{\text{bulk}} \Xi = 0$ is a linear, timeless equation, the bulk analogue of the Wheeler–DeWitt constraint. Physical time does not exist there. It appears at the boundary as the irreversible direction of the phase $\arg(\xi/\eta)$, generated by the breathing dynamics of Section 3.12.

3.8 The projected Noether balance

In the bulk $\mathcal{M}^5$, the single conservation law $\partial_I J^I = 0$ is a global statement. After projection $\Pi : \mathcal{M}^5 \rightarrow \mathbb{R}^4$, it acquires a local directional form at the boundary. Since Π is not isometric, the longitudinal component J^τ and the transverse components J^μ are mapped to distinct boundary sectors. The projected conservation law becomes:

$$\partial_\xi J^\xi = -\nabla_\eta J^\eta. \quad (9)$$

This is the fundamental boundary balance equation. It does not say the two sides vanish separately. It says: whatever the longitudinal channel ξ gains in coherent inflow, the transverse channel η must compensate with radiative outflow, and conversely.

Call ξ the *longitudinal channel*: coherent inflow, curvature, mass, gravitational binding. Call η the *transverse channel*: phase outflow, radiation, electromagnetic diffusion. Gravity and electromagnetism are not two forces unified by ansatz. They are the two compensating faces of one Noether current.

3.9 The local nonlinear Dirac equation as closure of the balance

In the bulk, the Majorana dynamics is linear and timeless: $\mathcal{D}_{\text{bulk}}\Xi = 0$. The boundary field $\Psi = \Pi\Xi$ does not inherit a closed linear equation, because $\ker\Pi \neq \{0\}$.

The boundary equation is the Euler–Lagrange equation of the projected boundary action induced by the bulk action on $\mathcal{M}^5$. The bulk Lagrangian $\mathcal{L}_{\text{QGT}} = \bar{\Xi}\gamma^A D_A \Xi - (4\alpha)^{-1} F_{AB} F^{AB} + \mathcal{L}_{\text{curv}}$ contains a curvature coupling term whose coefficients are fixed by three *bulk coherence conditions*, the conditions on the zero mode in the bulk that ensure the field Ξ is coherent on $S^1(\tau)$:

$$(2\alpha\eta)^2 = \varphi^2 \cdot 10^{-4}, \quad (2\alpha\xi)^2 = e/2, \quad \sqrt{\xi\eta}/3 \approx \pi. \quad (10)$$

The first two fix the transverse and longitudinal amplitudes of the temporal connection. The third, $\sqrt{\xi\eta} \approx 3\pi$, is the holonomic closure condition: π enters here not as a bulk curvature but as the measure of cyclic phase closure at the boundary of $S^1(\tau)$, the minimal observable half-cycle. These are not free parameters: they fix the local Dirac amplitudes and the boundary metric normalisation. They are not imposed as independent constraints on the final invariant $\mathcal{K}_\tau = \xi^* \eta^* = \alpha^{-1}$, which is fixed instead by the rank-5 spectral pole (Section 4). Their role is to define the local normalisation of the nonlinear Dirac closure, not to redetermine α . Note on e : the Euler number $e = 2.718\dots$ is not in $\mathbb{Z}[\sqrt{5}]$ and enters condition (10) as the near-identity $e \approx (\sqrt{2})^3 = 2\sqrt{2}$ (error 4%), which is the third level of the spinorial lever hierarchy $\{2^{1/4}, 2^{1/2}, 2^{3/4}, 2\}$. The condition $(2\alpha\xi)^2 = e/2$ is a structural identification of the Lagrangian coefficient; it fixes the structure of the nonlinear Dirac equation but α is determined independently by the spectral pole of the filter $H_\alpha(z)$ and does not depend on this approximation.

Remark 2 (Spinorial lever hierarchy and braid structure). The coherence condition $(2\alpha\xi)^2 = e/2$ belongs to a spinorial lever hierarchy $\{2^{k/4}\}_{k=1}^4$ that admits a braid-theoretic interpretation via the Jones representation of B_4 . Since this structure is not used in the derivation of α^{-1} , the full discussion is deferred to separate work.

Varying the action with respect to $\bar{\Psi}$ and applying the Euler–Lagrange equation gives:

- the kinetic term $\bar{\Xi}\gamma^\mu D_\mu \Xi$ produces $\gamma^\mu D_\mu \Psi$;
- condition $(2\alpha\xi)^2 = e/2$ fixes ξ as the longitudinal amplitude; variation of $\alpha\xi\bar{\Psi}\Psi$ gives $+\alpha\xi$;
- condition $(2\alpha\eta)^2 = \varphi^2 \cdot 10^{-4}$ fixes η as the transverse amplitude with chiral structure; variation of $i\alpha\eta\bar{\Psi}\gamma^5\Psi$ gives $+i\alpha\eta\gamma^5$.

The γ^5 on the transverse term is not arbitrary: η carries $U(1)$ phase (transverse, imaginary, radiative) while ξ carries $SU(2)$ curvature (longitudinal, real, massive). Their geometric distinction is encoded in the Clifford algebra by the chiral projectors $P_{\varphi\pm} = \frac{1}{2}(1 \pm \Gamma_6)$ of Section 3. The projection residue $\ker\Pi$ returns to the boundary as the functional gradient of the kernel entropy, giving the full boundary equation:

$$(\gamma^\mu D_\mu + \alpha\xi + i\alpha\eta\gamma^5)\Psi = \frac{\delta\mathcal{S}_{\text{ker}}[\Psi]}{\delta\bar{\Psi}}. \quad (11)$$

Here $\mathcal{S}_{\text{ker}}[\Psi] := -\text{tr}_{\ker\Pi}(\rho_\Psi \log \rho_\Psi)$ is the von Neumann entropy of the kernel sector in the state induced by $\Psi = \Pi\Xi$. The state ρ_Ψ is the reduced density matrix obtained by tracing out the boundary degrees of freedom from the pure state $|\Xi\rangle\langle\Xi|$:

$$\rho_\Psi := \text{tr}_{\text{Im}\Pi}(|\Xi\rangle\langle\Xi|) = \sum_{k=0}^{\infty} w_k(\Psi) |k\rangle_{\text{ker}}\langle k|_{\text{ker}}, \quad (12)$$

where $w_k(\Psi) \propto |\langle k|\Pi^\dagger\Psi\rangle|^2$ are the kernel-sector weights induced by Ψ . For the boundary return calculation we restrict $\mathcal{S}_{\text{ker}}$ to the rank-5 kernel-return channel induced by the retained boundary modes, with golden weights $w_k \propto \varphi^{-k}$, $k = 0, \dots, 4$. This is not the full kernel Hilbert space, but the finite

sector that couples back to the observable rank-5 boundary. Its gradient $V'(|\Psi|) := \delta \mathcal{S}_{\text{ker}} / \delta |\Psi|$ is the return tension, the force with which the kernel pulls the boundary field back toward the bulk coherence it cannot fully carry. This is the local closure of the projected balance (9). The linear limit recovers the generalised chiral Dirac equation; the non-linear term is the return tension of the zero mode that did not cross the interface.

Remark 3 (The kernel as Hilbert quadrature of the boundary). The structure of the projection Π admits a transparent signal-theoretic reading that sharpens the physical meaning of $\text{ker} \Pi$ and of the nonlinear return term $\delta \mathcal{S}_{\text{ker}} / \delta \bar{\Psi}$.

The projection $\Pi = \text{Re}[\cdot]$ extracts the real (longitudinal) part of the bulk Majorana field on $S^1(\tau)$. The boundary field is therefore the real part of an analytic signal:

$$\Psi_{\text{bdy}} = \text{Re}(\Xi), \quad \Xi_{\text{ker}} = -i \mathcal{H}[\Psi_{\text{bdy}}], \quad (13)$$

where $\mathcal{H}$ is the Hilbert transform on the compact fibre $S^1(\tau)$. The kernel spinor Ξ_{ker} is therefore *not* an arbitrary hidden sector: it is the Hilbert quadrature of the observable boundary field.

This identification has a direct physical consequence. The nonlinear return term $\delta \mathcal{S}_{\text{ker}} / \delta \bar{\Psi}$ is the local back-action of this unobserved quadrature on the projected boundary dynamics: the imaginary part of the analytic signal cannot propagate to the boundary, but it acts on it as a restoring tension, driving the boundary field toward the stable operating point Ψ_* . In this sense:

$$\begin{aligned} \text{ker} \Pi &= \text{Hilbert quadrature of the boundary} \\ \implies \frac{\delta \mathcal{S}_{\text{ker}}}{\delta \bar{\Psi}} &= \text{back-action of the hidden quadrature.} \end{aligned}$$

The pole α^{-1} of the boundary filter $H_\alpha(z)$ is the stable operating point precisely because the kernel in quadrature closes the boundary dynamics dynamically: the system cannot settle at any other value without violating the conservation $\xi \eta = \alpha^{-1}$.

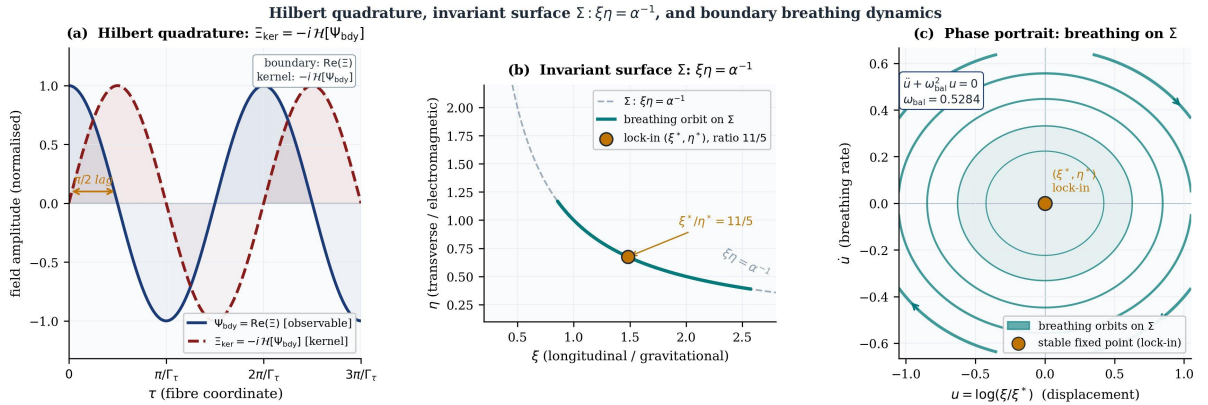


Figure 1: Hilbert quadrature, invariant surface, and boundary breathing dynamics. (a) The observable boundary field $\Psi_{\text{bdy}} = \text{Re}(\Xi)$ and the kernel quadrature $\Xi_{\text{ker}} = -i \mathcal{H}[\Psi_{\text{bdy}}]$ form a Hilbert pair on the compact fibre $S^1(\tau)$: the boundary sees the real part of the bulk Majorana field; the kernel carries its Hilbert transform. (b) The two chiral channels (ξ, η) live on the invariant hyperbolic surface $\Sigma: \xi \eta = \alpha^{-1}$. The breathing orbit (teal) oscillates on Σ around the lock-in point (ξ^*, η^*) with ratio $\xi^*/\eta^* = 11/5$. (c) Phase portrait of the breathing equation $\ddot{u} + \omega_{\text{bal}}^2 u = 0$, $u := \log(\xi/\xi^*)$, $\omega_{\text{bal}} = 0.5284$. The stable fixed point at the origin is the lock-in; the closed orbits are the boundary breathing modes. The fine-structure constant α^{-1} is the conserved invariant of this projected exchange, not an externally inserted parameter.

3.10 The boundary code subspace and the stable operating point

The nonlinear boundary equation (11) has a stable fixed point Ψ_* satisfying $\mathcal{F}[\Psi_*] = 0$. Linearising around Ψ_* , the tangent operator $\mathcal{L}_* := \delta\mathcal{F}/\delta\Psi|_{\Psi_*}$ has exactly five eigenmodes $v_0, \dots, v_4$: the *boundary code subspace* of the rank-5 projection. These are the modes that survive the finite-rank projection. The rational transfer function $H_\alpha(z)$ of Section 4 is the spectral response of this code subspace; its pole $z_{\text{pole}} = \alpha^{-1}$ is the stability threshold of the observable boundary.

The dynamics of the two channels at this fixed point is:

$$\dot{\xi} = -\Gamma_\tau \xi, \quad \dot{\eta} = +\Gamma_\tau \eta, \quad (14)$$

or equivalently, in the logarithmic coordinates $u = \log \xi$, $v = \log \eta$: $\dot{u} = -\Gamma_\tau$, $\dot{v} = +\Gamma_\tau$. These are not postulated: they are the unique dynamics of the Pincherle dilatation operator $\mathcal{D} = x\partial_x$ on the two autosimilar channels at rate $\Gamma_\tau = \varphi/\sqrt{5}$, projected onto the boundary. In the logarithmic plane the flow is uniform; in the (ξ, η) plane it is a *hyperbolic rotation* on the surface Σ . The product $\xi\eta$ is conserved exactly:

$$\frac{d}{dt}(\xi\eta) = \dot{\xi}\eta + \xi\dot{\eta} = -\Gamma_\tau\xi\eta + \xi\Gamma_\tau\eta = 0.$$

The surface invariant:

$$\boxed{\mathcal{K}_\tau = \xi\eta = \alpha^{-1}} \quad (15)$$

is therefore preserved at every instant by the hyperbolic rotation, not only at the fixed point. The coherent channel ξ decays at rate Γ_τ (amplitude decreasing); the radiative channel η grows at rate Γ_τ (amplitude increasing); their product is the Noether charge of the temporal fibre, permanently locked.

α as the *time quantum of the boundary*. The elementary phase quantum of the projected boundary is $\pi\alpha$: the minimal phase step resolvable at the rank-5 boundary, set by the resolution floor of the boundary metric (the same quantity that appears in the two-sided exclusion of the rank, Section 4). Numerically $\pi\alpha = \pi/\alpha_{\text{QGT}}^{-1} \approx 2.3 \times 10^{-2}$. This is the geometric resolution threshold of the boundary: the minimal phase separation below which two distinct bulk states project to the same boundary observable, not because of measurement precision but because their difference lives in $\ker \Pi$.

The reason this threshold is what it is comes from the entropic window of the rank-5 boundary:

$$S_\varphi = \frac{\varphi^4}{5} \approx 1.37 \text{ nat} \approx 1.98 \text{ bit}. \quad (16)$$

S_φ is precisely sufficient to support two independent binary distinctions and insufficient to support three. Two distinctions, and only two: the longitudinal channel ξ (curvature, mass, gravity) and the transverse channel η (phase, radiation, electromagnetism). This is the information-theoretic reason why the single Noether current $\partial_t J^t = 0$ produces exactly the two-channel balance $\partial_\xi J^\xi = -\nabla_\eta J^\eta$, and not a three-channel or single-channel decomposition. The resolution floor $\pi\alpha$ is the phase-space expression of this 2-bit capacity.

At the stable operating point $\xi^* \eta^* = \alpha^{-1}$:

$$\alpha = \frac{1}{\xi\eta}, \quad \pi\alpha = \frac{\pi}{\xi\eta}, \quad (17)$$

so $\alpha^{-1} = \xi\eta$ is the invariant of the balance, and $\pi\alpha$ is the elementary phase quantum, one half-cycle of the boundary oscillation in units of π . The Gaussian measure of the continuous autosimilar hierarchy on $S^1(\tau)$ ($\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$, the natural normalisation of the bulk) sets the unit. The rank-5 projection discretises that continuous measure; its stable operating point is α^{-1} . In this sense, α^{-1} is the time quantum of the boundary: the stable product of longitudinal inflow and transverse outflow that the single Noether current sustains under rank-5 projection.

The exchange (14) is the physical origin of time at the boundary. In the bulk, $\mathcal{D}_{\text{bulk}} \Xi = 0$ is timeless. At the boundary, the logarithmic phase $\theta_{\log} = \frac{1}{2} \log(\eta/\xi)$ advances irreversibly at rate Γ_τ : $\dot{\theta}_{\log} = \frac{1}{2}(\dot{\eta}/\eta - \dot{\xi}/\xi) = \Gamma_\tau$. That irreversible advance is what a boundary observer calls time.

3.11 The palindromic structure of the projection

The invariant $\xi\eta = \alpha^{-1}$ encodes a deeper symmetry. Normalise $\hat{\xi} := \sqrt{\alpha}\xi$, $\hat{\eta} := \sqrt{\alpha}\eta$, giving $\hat{\xi}\hat{\eta} = 1$. The ratio $x := \hat{\xi}/\hat{\eta}$ satisfies:

$$\lambda^2 - k\lambda + 1 = 0, \quad k = x + x^{-1}, \quad (18)$$

whose constant term $+1$ encodes $\hat{\xi}\hat{\eta} = 1$. This palindromic structure is the boundary image of Majorana self-conjugacy: charge conjugation maps $x \leftrightarrow x^{-1}$.

The Pincherle dilatation $D := \tau\partial_\tau$ generates the multiplicative structure of the bulk. Under projection, D compresses to $D_{\text{eff}} := \Pi D \Pi^*$, inheriting the palindromic symmetry:

$$D_{\text{eff}}^2 - kD_{\text{eff}} + 1 = 0. \quad (19)$$

All subsequent structural results follow from (19) together with the rank constraint.

3.12 The breathing equation as discrete Noether balance

The breathing equation is not postulated. It arises from the incompatibility of two structural closure conditions that the boundary must satisfy simultaneously after projection.

The three conditions (10) fix the bulk zero mode. After projection, the boundary amplitude $\mathcal{B} := |\Psi|^2$ must also satisfy two independent *spectral closure conditions*:

The *holonomic closure condition* arises from the $\text{SU}(2) \rightarrow \text{SO}(3)$ descent accumulated over three half-turns of the Majorana spinor on the boundary:

$$\mathcal{B}^{1/2} = 3\pi. \quad (20)$$

The *spectral closure condition* arises from the dimensional reduction $S^2 \times S^1(\tau) \rightarrow S^2$ at spectral curvature $\kappa = \sqrt{5}$:

$$\mathcal{B}^{1/3} = 2\sqrt{5} = C_{\text{bulk}}. \quad (21)$$

These two conditions are structurally incompatible: (20) gives $\mathcal{B} = 9\pi^2 \approx 88.83$, while (21) gives $\mathcal{B} = (2\sqrt{5})^3 = 40\sqrt{5} \approx 89.44$. They cannot be satisfied simultaneously because $\pi \notin \mathbb{Q}(\sqrt{5})$, the transcendental and algebraic scales live in different fields. The 0.7% gap between $9\pi^2$ and $40\sqrt{5}$ is not an inconsistency: it is the structural tolerance of the projection, the quantitative signature of the information lost when a volumetric bulk object ($\mathcal{B}^{1/3}$) is compressed onto an observable surface ($\mathcal{B}^{1/2}$).

The breathing equation is the map that propagates $\mathcal{B}$ from level k to $k+1$ under these two independent constraints:

$$\mathcal{B}_{k+1} = 2\mathcal{B}_k^{2/3}. \quad (22)$$

The exponent $2/3$ is forced by condition (21): the coherence density at level k maps as $\mathcal{B} \mapsto \mathcal{B}^{2/3}$, the ratio of boundary to bulk sphere dimensions under the spectral reduction. The prefactor 2 is forced by condition (20): the Majorana spinor requires 4π rotation to return to itself; the projection halves this to 2π , contributing the factor 2.

The unique positive fixed point is found from $\mathcal{B}^* = 2(\mathcal{B}^*)^{2/3}$:

$$\mathcal{B}^* = 2^3 = 8, \quad (23)$$

a global attractor with Lyapunov exponent $2/3 < 1$. At this fixed point, $(\mathcal{B}^*)^{1/3} = 2 = C_{\text{bulk}}/\sqrt{5}$: the boundary attractor and the bulk curvature are related by the spectral curvature $\kappa = \sqrt{5}$, closing the chain.

The value $\mathcal{B}^* = 8 = \dim \mathfrak{su}(3)_c$ is not a numerical accident: both are residues of the same rank-5 projection. We conjecture that the equality reflects a structural identity between boundary amplitude stability and colour confinement.

The breathing back-action correction f_{breath} in Section 5 originates from the tensorial structure of α : since the discrete breathing map samples the boundary at rate $\varphi^2/10$, any measurement of α samples the step-averaged norm of the tensor $\alpha = (\alpha_{\parallel}, \alpha_{\perp})$. The algebraic direction $(1 + \sqrt{5}) = 2\varphi$ is fixed by the exact identity $2\varphi/(5 + 3\sqrt{5}) = (5 - \sqrt{5})/10$ in $\mathbb{Z}[\sqrt{5}]$; the absolute scale 10^{-7} is fixed by the confined mode $n_{\text{conf}} = 7$ as shown in Section 5.

Remark 4 (Kernel, lossiness, and the problem of time). The non-invertibility of Π is the projection-theoretic signature of bulk-boundary entanglement: $\ker \Pi$ contains precisely the degrees of freedom that are entangled with the boundary but cannot be reconstructed from boundary data alone. The projection takes the real part of the analytic signal on $S^1(\tau)$: $\Psi_{\text{bdy}} = \Re(\Xi)$ and $\Xi_{\ker} = -i\mathcal{H}[\Psi_{\text{bdy}}]$, where $\mathcal{H}$ is the Hilbert transform on $S^1(\tau)$. Observable physics is $\Re(\Psi)$; the kernel is its Hilbert quadrature: physically active, not directly accessible. The constants of $\mathbb{R}^4$ are stability conditions of this lossy boundary. In the bulk, $\mathcal{D}_{\text{bulk}}\Xi = 0$ is linear and timeless, the bulk analogue of the Wheeler–DeWitt constraint. Physical time appears at the boundary as the irreversible direction of the phase $\arg(\xi/\eta)$, generated by the breathing of the return tension $\delta\mathcal{S}_{\ker}/\delta\bar{\Psi}$. The internal structure of $\ker \Pi$ carries a deeper arithmetic signature: the Heegner index $163 = \lfloor \alpha^{-1} \rfloor + 2 \times 13 = 137 + 26$ connects the integer part of α^{-1} to the electroweak denominator $c_{\text{EW}} = 13$ via the largest Heegner number. The Gaussian normalisation of the bulk, the Heegner index of the kernel, and the phase quantum $\pi\alpha$ of the boundary are three faces of the same quadratic-form arithmetic; the complete derivation is deferred.

4 How the projection rank and the value of α are determined

The palindromic operator (19) carries all the structural information about the projection, but it does not yet tell us the rank. The rank is determined by a separate question: how many independent modes can Π support on $S^1(\tau)$ before the singular spectrum falls below the resolution threshold? Two independent derivations (one from the spectral curvature, one from the Majorana recursion) converge on the same answer.

4.1 How many modes the projection can support

The palindromic operator (19) forces the characteristic eigenvalue $\varphi = (1 + \sqrt{5})/2$ and spectral curvature $\kappa = 2(\varphi - \frac{1}{2}) = \sqrt{5}$, independently of the rank. Mode n of Π contributes singular value $\sigma_n = \kappa^{W-n}$ to the boundary observable sector, where W is the winding capacity: the largest n above the unit threshold $\sigma_n \geq 1$. The Fibonacci condition $F_W = W$ (derived independently from the Majorana recursion in the next subsection) has the unique non-trivial solution $W = 5$. The spectral threshold confirms:

$$\sigma_5 = \kappa^0 = 1 \geq 1, \quad \sigma_6 = \kappa^{-1} = 5^{-1/2} < 1. \quad (24)$$

Exactly five modes lie above threshold; the rank of the projection is $r = W = 5$. The rank sets the information capacity of Π : exactly five modes survive to the boundary; all modes with $n > 5$, including the confined mode $n = 7$, reside in $\ker \Pi$ and carry the bulk entropy inaccessible to the boundary observer.

4.2 The unique self-consistent mode number

The Majorana condition $\Xi = C\bar{\Xi}$ means the field is its own charge conjugate (for experimental realizations of Majorana-like quasi-particles in quantum simulators see [3]). Charge conjugation C commutes with the free Hamiltonian $\hat{H}$, so the mode eigenspaces are mutually orthogonal: $\langle \Xi_n, \Xi_m \rangle = 0$ for $n \neq m$. The transport of Ξ on $S^1(\tau)$ under Γ_τ satisfies a general second-order recurrence $\Xi_{n+2} = a\Xi_{n+1} + b\Xi_n$. The Majorana self-conjugacy requires charge-conjugation symmetry between the two branches, forcing $a = b$. That $a = b = 1$ specifically, not merely $a = b$, follows from three independent conditions:

Proposition 3 ($a = b = 1$: three-condition uniqueness). *The recurrence coefficients are $a = b = 1$, uniquely, because:*

- (i) Holonomy integrality. *Each mode Ξ_n acquires holonomy $e^{i\pi/n}$ on $S^1(\tau)$. The Majorana reality $C\bar{\Xi}_n = \Xi_n$ forces $a^2 + b^2 \in \mathbb{Z}$; with $a = b$ this gives $2a^2 \in \mathbb{Z}$, so $a \in \frac{1}{\sqrt{2}}\mathbb{Z}$. Positivity and integrality together force $a \in \mathbb{Z}_{\geq 1}$.*

- (ii) Minimality (no redundant lift). A non-invertible projection Π must not identify distinct bulk modes with the same boundary state. For positive a, b this is equivalent to $ab \leq 1$. Combined with $a = b$ and $a \geq 1$: $a = 1$.
- (iii) Unit boundary normalisation. The Jackiw–Rebbi edge modes at the domain walls normalise to $|O_1|^2 = |O_2|^2 = 1$ by the boundary metric condition. With $b = 1$ (from (i)–(ii)) and $a = 1$, the recursion is uniquely $|O_{n+2}|^2 = |O_{n+1}|^2 + |O_n|^2$ with unit initial conditions.

No other pair (a, b) with $a, b > 0$ satisfies all three simultaneously.

Proof. From (i): Majorana forces $a = b$, so the recursion is $|O_{n+2}|^2 = a(|O_{n+1}|^2 + |O_n|^2)$. Holonomy integrality on $S^1(\tau)$ forces $a^2 + b^2 = 2a^2 \in \mathbb{Z}_{>0}$, so $a \in \mathbb{Z}_{\geq 1}$. From (ii): minimality requires $ab = a^2 \leq 1$ with $a \geq 1$, forcing $a = 1$. From (iii): $|O_1|^2 = |O_2|^2 = 1$ confirms $b = a = 1$. Uniqueness: any $a > 1$ violates (ii); any $a < 1$ fails integrality; $a \neq b$ violates (i). $\square$

Remark 5 (The Fibonacci transfer matrix). The transport operator on $S^1(\tau)$ that generates the Fibonacci recursion is the companion matrix:

$$T_F = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} \Xi_{n+2} \\ \Xi_{n+1} \end{pmatrix} = T_F \begin{pmatrix} \Xi_{n+1} \\ \Xi_n \end{pmatrix}. \quad (25)$$

Its entries $(1, 1, 1, 0)$ follow from Proposition 3 without freedom. Its characteristic polynomial $\lambda^2 - \lambda - 1 = 0$ has roots φ and $-1/\varphi$, the eigenvalues of γ^6 in scale space. The Fibonacci recursion is therefore the spectral algebra of the Majorana chiral projector on $S^1(\tau)$, not an assumption.

Taking the squared norm and using orthogonality:

$$|O_{n+2}|^2 = |O_{n+1}|^2 + |O_n|^2, \quad |O_1|^2 = |O_2|^2 = 1. \quad (26)$$

which is the Fibonacci recursion with $|O_n|^2 = F_n$.

The equation $F_n = n$ admits exactly one non-trivial solution. Direct verification: $F_1 = 1 = 1$ (trivial); $F_2 = 1 \neq 2$; $F_3 = 2 \neq 3$; $F_4 = 3 \neq 4$; $F_5 = 5 = 5$ (non-trivial). For $n \geq 6$: $F_6 = 8 > 6$ provides the base, and if $F_k > k$ and $F_{k-1} > k-1$, then $F_{k+1} > 2k-1 \geq k+1$, so by induction $F_n > n$ for all $n \geq 6$. The unique non-trivial solution is

$$F_{n_*} = n_* = r = 5. \quad (27)$$

Lemma 1 (Rank-locking: geometric and algebraic derivations converge). The rank $r = 5$ is determined by two independent derivations that approach it from opposite directions:

- (i) Geometric (spectral threshold). The palindromic operator forces spectral curvature $\kappa = \sqrt{5}$. Mode n survives the boundary threshold iff $\kappa^{W-n} \geq 1$. The winding capacity is $W = 5$: $\sigma_5 = \kappa^0 = 1 \geq 1$ and $\sigma_6 = \kappa^{-1} < 1$. Exactly five modes survive; $r = 5$.
- (ii) Algebraic (Fibonacci fixed point). Majorana self-conjugacy forces $|O_n|^2 = F_n$. The fixed-point condition $F_{n_*} = n_*$ has unique non-trivial solution $n_* = 5$. The Fibonacci occupancy closes on itself at $n = 5$: $W = 5$.

Both derivations give $r = 5$. Their agreement is the rank-locking: the unique integer at which the spectral geometry of the projection and the algebra of the Majorana field coincide. This is not a coincidence or a separate axiom, it is the structural fingerprint of a continuous bulk generating a discrete boundary through a non-invertible projection.

The threshold check (24) is a verification, not a definition.

Remark 6 (Admissible mode catalogue). The condition $\pi(n) \leq r$, where $\pi(n)$ is the prime-counting function measuring irreducible angular complexity, selects $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$. The mode $n_{\text{conf}} = 7 \in \mathcal{N}_{\text{adm}}$ is confined: it does not factorise within the Fibonacci basis $\{2, 3, 5\}$ and lives in $\ker \Pi$.

4.3 The unique transfer function of the projection

We now know the rank ($r = 5$), the palindromic structure of D_{eff} , and the Noether normalisation ($\mathcal{K}_\tau = \alpha^{-1}$). These three constraints together force the rational transfer function of the projection to be unique. There is no freedom left.

Given the palindromic operator (19), rank $r = 5$, a single spectral pole, and the log-cumulant basis $v_m(x) \propto (\log x)^m$ ($m = 0, \dots, 4$) adapted to $\{\varphi^k\}$, the rational transfer function of the projection is unique:

$$H_\lambda(z) = \frac{P_4(z)}{1 - \lambda z}, \quad P_4(z) = \sum_{m=0}^4 (\log \varphi)^m z^m, \quad (28)$$

where λ^{-1} is the spectral pole, to be determined. No value of λ is assumed; the pole location is derived next. The proof that no other rational transfer function satisfies all four constraints simultaneously is given in Appendix B.

4.4 The spectral pole and its algebraic structure

The filter $H_\lambda(z) = P_4(z)/(1 - \lambda z)$ has a single pole. Where is it? The Fibonacci fixed point identifies the lattice element $x_* = 5 + 3\sqrt{5} \in \mathbb{Z}[\sqrt{5}]$. The spectral pole is:

$$\alpha_{\text{QGT}}^{-1} = (5 + 3\sqrt{5})^2 = 70 + 30\sqrt{5} \approx 137.082. \quad (29)$$

The derivation chain is:

$$S^1(\tau) \Rightarrow \partial_t J^I = 0 \Rightarrow \Xi = C\bar{\Xi}, (\gamma^6)^2 = 5 \Rightarrow r = 5 \Rightarrow H_\alpha(z) \Rightarrow \alpha_{\text{QGT}}^{-1} = 70 + 30\sqrt{5}. \quad (30)$$

No number is inserted at any step. This chain is a *mathematical theorem*: each arrow is either a direct computation or a previously proved proposition (Propositions 1–3, Lemma 1, Appendix B). The symbol α is assigned only at the last arrow, after its value is derived. It does not appear as input anywhere in the chain.

The Fibonacci fixed point identifies the lattice element $x_* = 5 + 3\sqrt{5} \in \mathbb{Z}[\sqrt{5}]$ with coordinates $(F_5, F_4) = (5, 3)$. This determines the spectral pole: $\lambda^{-1} = (5 + 3\sqrt{5})^2 = 70 + 30\sqrt{5}$. We now name the projection constant $\alpha_{\text{QGT}} := \lambda$, so $\alpha_{\text{QGT}}^{-1} = 70 + 30\sqrt{5}$, and henceforth write $H_\alpha(z) := H_\lambda(z)$. The symbol α is assigned *after* its value is derived, not before. The rank-5 projection Π induces a truncated SVD on the boundary; its action on $\mathbb{Z}[\sqrt{5}]$ in the basis $\{1, \sqrt{5}\}$ is encoded by the 2×2 integer matrix $\mathbf{M}$, which is the *root matrix* of the boundary: its squared $\mathbb{Z}[\sqrt{5}]$ -norm gives α^{-1} , and its off-diagonal entry encodes the gauge-class count $|\mathcal{C}_{\text{obs}}| = 3$ that determines the electroweak sector:

$$\mathbf{M} = \begin{pmatrix} F_5 & \kappa^2 F_4 \\ F_4 & F_5 \end{pmatrix} = \begin{pmatrix} 5 & 15 \\ 3 & 5 \end{pmatrix}, \quad \det \mathbf{M} = -n_{\text{min}}^2 r = -20, \quad \|\mathbf{M}^2\|_{\mathbb{Z}[\sqrt{5}]} = \alpha_{\text{QGT}}^{-1}. \quad (31)$$

M as regular representation. The matrix $\mathbf{M}$ is not an auxiliary choice. It is the unique 2×2 integer matrix representing multiplication by $x = 5 + 3\sqrt{5}$ in the $\mathbb{Z}[\sqrt{5}]$ -module with basis $\{1, \sqrt{5}\}$:

$$x \cdot 1 = 5 \cdot 1 + 3\sqrt{5}, \quad x \cdot \sqrt{5} = 15 \cdot 1 + 5\sqrt{5},$$

so $\mathbf{M} = \text{reg}(5 + 3\sqrt{5})$ is forced. Its square and $\mathbb{Z}[\sqrt{5}]$ -norm:

$$\mathbf{M}^2 = \begin{pmatrix} 70 & 150 \\ 30 & 70 \end{pmatrix}, \quad \|\mathbf{M}^2\|_{\mathbb{Z}[\sqrt{5}]} = (5 + 3\sqrt{5})^2 = 70 + 30\sqrt{5} = \alpha_{\text{QGT}}^{-1}. \quad (32)$$

The pole is the squared $\mathbb{Z}[\sqrt{5}]$ -norm of the regular representation of the Fibonacci fixed-point element.

The entries are $(r, \kappa^2 |\mathcal{C}_{\text{obs}}|, |\mathcal{C}_{\text{obs}}|, r) = (5, 15, 3, 5)$: the projection rank on the diagonal, the spectral curvature $\kappa^2 = 5$ times the gauge-class count $|\mathcal{C}_{\text{obs}}| = 3$ off-diagonal. The negative determinant reflects

Majorana self-conjugacy ($\Xi = C\bar{\Xi}$ implies zero net bulk charge). The sectional curvatures of the two chiral sectors satisfy the closure:

$$\frac{c_{\text{EW}} + n_{\text{conf}}}{\kappa^2} = \frac{13 + 7}{5} = 4 = \dim \mathbb{R}^4, \quad (33)$$

where $c_{\text{EW}} = 2r + |\mathcal{C}_{\text{obs}}| = 13$ and $n_{\text{conf}} = 7$: the electroweak and colour sectors together cover exactly the four observable dimensions. The gauge-sector consistency checks in Section 6 arise from the scalar contractions of $\mathbf{M}$: $\|\mathbf{M}^2\|_{\mathbb{Z}[\sqrt{5}]}$ gives α^{-1} and the diagonal ratio of the gauge tensor built from the same Fibonacci entries gives $\sin^2 \theta_W = 3/13$.

5 Two corrections from the boundary dynamics

5.1 Bare pole and observed pole

The projection filter $H_\alpha(z)$ has a static spectral pole at $\alpha_{\text{QGT}}^{-1} = 70 + 30\sqrt{5} \approx 137.082$. This is the *bare projection pole*: the value of α^{-1} in the absence of boundary dynamics, computed at the symmetric point $\xi = \eta$. It is not the measured value.

The *observed pole* is the cycle-averaged value accessible to a low-energy boundary observer. Two structural effects separate the observed from the bare: (i) the holonomy residue of the confined mode $n = 7$, a first-order correction exact in $\mathbb{Z}[\sqrt{5}][\pi]$; and (ii) the breathing back-action, a second-order multiplicative renormalisation exact in $\mathbb{Z}[\sqrt{5}]$. Neither contains a fitted parameter. The analogy with quantum field theory is instructive but limited: the bare pole is the tree-level quantity; the two corrections are *structural*, they arise from the projection geometry before any QED loop diagram is computed. They do not replace the perturbative renormalisation group: they determine the boundary input value $\alpha^{-1} \approx 137.036$ at the atomic lock-in scale, from which QED running then takes over. The form of both corrections is fixed by the projection, not by fitting.

The static pole $\alpha_{\text{QGT}}^{-1} = 70 + 30\sqrt{5} \approx 137.082$ is the value of α^{-1} at the symmetric operating point $\xi = \eta$. But the physical boundary is not symmetric: the lock-in selects $\xi^*/\eta^* = 11/5$, and the confined mode $n = 7$ is not visible at the boundary but bleeds through the electroweak channels. Two structural effects shift the static pole toward the measured value. Neither requires a fitted parameter; both have their algebraic form fixed in $\mathbb{Z}[\sqrt{5}]$. We compute them in sequence.

5.1.1 Holonomy correction

Lemma 2 ($n = 7$ unique first-order contribution; $c_{\text{EW}} = 13$ forced). *Among all modes $n \in \mathcal{N}_{\text{adm}}$:*

- (a) *$n = 7$ is the unique mode in $\ker \Pi$ that contributes to the first-order correction of the static pole. The modes $n \in \{2, 3, 5\}$ are in $\text{Im} \Pi$ and generate no kernel residue. The mode $n = 11$ is not in $\ker \Pi$: it is the domain-wall saturation mode (DW_2) and produces no holonomy residue at the boundary. In the sensitivity table, replacing $n = 7$ with $n = 11$ is included as a counterfactual variant, not as an admissible first-order correction.*
- (b) *The bleed-through multiplicity is $c_{\text{EW}} = 2r + |\mathcal{C}_{\text{obs}}| = 2 \cdot 5 + 3 = 13$, the same denominator forced by Section 6 for $\sin^2 \theta_W$. Without the Majorana constraint, a naive doubled bookkeeping would count $2c_{\text{EW}} = 26$ channel copies. The Majorana self-conjugacy $\Xi = C\bar{\Xi}$ identifies each channel with its conjugate, reducing the physical multiplicity from 26 to $c_{\text{EW}} = 13$.*

The correction $13\epsilon_7$ and the mixing angle $3/13$ are two projections of the same rank-5 object.

Proof. (a) $n = 7 \in \ker \Pi$ and $n = 7 \notin \text{Im} \Pi$. The admissible mode $n = 7$ does not factorise within the Fibonacci vocabulary $\{2, 3, 5\}$ of the lock-in, so it cannot appear as a Lissajous class of the observable boundary. Equivalently, its SVD weight is $\sigma_{n=7} = 0$: mode 7 does not factorise within the Fibonacci basis $\{2, 3, 5\}$ and carries no boundary projection weight. Hence $n = 7$ is confined: $7 \in \ker \Pi$. The modes $n \in \{2, 3, 5\}$ are in $\text{Im} \Pi$ and generate no residue.

(b) $n = 11$ does not contribute. Mode $n = 11$ is the domain-wall saturation mode: the outer boundary of the admissible spectrum $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$. It is in $\mathcal{N}_{\text{adm}}$ but is not in $\ker \Pi$: it defines the outer boundary of the observable region, not a confined holonomy sector. Since $n = 11 \notin \ker \Pi$, it produces no holonomy residue at the observable boundary, and its contribution to α^{-1} is zero to all orders. Its inclusion in the sensitivity table as a counterfactual variant (row $n = 11$, $c = 13$, multiplicative) demonstrates that the physical correction is unique: replacing $n = 7$ with $n = 11$ would give ≈ 137.122 , off by 8.6×10^{-2} .

(c) $c_{\text{EW}} = 13$ forced. The electroweak denominator is $c_{\text{EW}} = 2r + |\mathcal{C}_{\text{obs}}| = 2 \cdot 5 + 3 = 13$. The two terms have distinct origins: $2r = 10$ counts the η - and ξ -channel copies of the $r = 5$ boundary modes; $|\mathcal{C}_{\text{obs}}| = 3$ counts the observable gauge classes (W^+ , W^- , Z^0 , forced by Section 6). Without the Majorana constraint, the naive doubled bookkeeping would give $2c_{\text{EW}} = 26$; the self-conjugacy $\Xi = C\bar{\Xi}$ identifies conjugate pairs, reducing $26 \rightarrow 13$. The same $c_{\text{EW}} = 13$ appears as the denominator of $\sin^2 \theta_W = 3/13$: both are contractions of the same rank-5 Fibonacci matrix $\mathbf{M}$ (derived in Section 6). $\square$

The confined mode $n_{\text{conf}} = 7 \in \ker \Pi$ bleeds through $c_{\text{EW}} = 2r + |\mathcal{C}_{\text{obs}}| = 13$ electroweak channels ($|\mathcal{C}_{\text{obs}}| = F_{n_s-1} = 3$). Mode $n = 7$ winds on $S^1(\tau)$ with holonomy angle $\pi/7$; the projection Π cannot absorb this winding (since $7 \in \ker \Pi$), so the amplitude $\kappa \cdot (\pi/7) = \sqrt{5}\pi/7$ bleeds through the electroweak channels as a boundary residue, reduced by the unit contribution of the base mode. Its holonomy residue at the boundary is:

$$\varepsilon_7 = \frac{\sqrt{5}\pi}{7} - 1 \in \mathbb{Z}[\sqrt{5}][\pi]. \quad (34)$$

This correction shifts α_{QGT}^{-1} to within 4.4×10^{-5} of the measured value.

Lemma 3 (The unit subtraction). *The subtracted 1 in (34) is the holonomy of one closed winding of the base mode on $S^1(\tau)$: a full unit winding lies in $\text{Im} \Pi$ and is absorbed by the cyclic readout, hence it is invisible at the boundary. The observable bleed-through is the excess over that closed unit; the subtraction is a structural unit, not an adjustable offset. The same rule applies uniformly to every counterfactual mode of Table 2: $\varepsilon_n = \sqrt{5}\pi/n - 1$.* $\square$

5.1.2 Breathing back-action

The fine-structure constant is not a scalar on the oscillating boundary. The exchange dynamics (14) show that $\xi(t)$ and $\eta(t)$ undergo a hyperbolic rotation on Σ : $\xi\eta = \alpha^{-1}$ with characteristic rate $\Gamma_\tau = \varphi/\sqrt{5}$. The anisotropy of the lock-in point $\xi^*/\eta^* = n_{\text{sat}}/r = 11/5$ modulates the effective rates at which the two channels present to a low-energy observer:

$$\omega_\xi = \Gamma_\tau \cdot \frac{n_*}{n_{\text{sat}}} = \frac{\varphi}{\sqrt{5}} \cdot \frac{5}{11}, \quad \omega_\eta = \Gamma_\tau \cdot \frac{n_{\text{sat}}}{n_*} = \frac{\varphi}{\sqrt{5}} \cdot \frac{11}{5}, \quad (35)$$

where $n_* = 5$ is the lock-in mode and $n_{\text{sat}} = 11 = \max \mathcal{N}_{\text{adm}}$ is the saturation mode. Their invariant product is $\omega_\xi \cdot \omega_\eta = \Gamma_\tau^2 = \varphi^2/5$, the square of the hyperbolic rotation rate. Any low-energy QED measurement of α samples the step-averaged norm of the two-component tensor $\alpha = \alpha^{(G)}(\xi, \eta)/\sqrt{\mathcal{K}}$, not the static pole.

The form $1/(1 - f_{\text{breath}})$ rather than $\alpha^{-1} + f_{\text{breath}}$ follows from the cycle average of α^{-1} over the breathing orbit. The boundary field $\alpha^{-1}(t) = \alpha_0^{-1}/(1 - f_{\text{breath}} \sin^2 \omega t)$ oscillates on the hyperbolic surface Σ : $\xi\eta = \alpha^{-1}$. Its time average is:

$$\langle \alpha^{-1} \rangle_{\text{cycle}} = \frac{\alpha_0^{-1}}{1 - f_{\text{breath}}} + O(f_{\text{breath}}^2), \quad (36)$$

where $\alpha_0^{-1} = \alpha_{\text{QGT}}^{-1} - c_{\text{EW}}\varepsilon_7$ is the holonomy-corrected static pole. This is the standard result for the geometric mean of an oscillating denominator: the multiplicative form $(1 - f)^{-1}$ is the first-order expansion of the cycle average, not an independent choice.

The absolute scale of the back-action is determined by the spectral structure of the projection. The back-action of one breathing cycle at observer level $k_{\text{obs}} = n_{\text{min}} = 2$ is the invariant product suppressed by the scale hierarchy $\mathcal{S}^{k_{\text{obs}}(1+n_{\text{min}})} = 10^{2 \times 3} = 10^6$ and renormalised by the Majorana coupling $\varphi = \kappa \Gamma_\tau$, the product of the conserved self-conjugacy norm $\kappa = \sqrt{5}$ (the invariant $\varphi + \varphi^{-1} = \sqrt{5}$ of the Majorana pair, the same κ that scales the holonomy amplitude feeding (34)) and the channel's native rate $\Gamma_\tau = \varphi/\sqrt{5}$: the pair-product rate $\omega_\xi \omega_\eta$ is referred to the single self-conjugate channel per unit of its own breathing rate. Both factors are fixed upstream; no new normalisation enters:

$$f_{\text{breath}} = \frac{\omega_\xi \cdot \omega_\eta}{\varphi \cdot \mathcal{S}^{k_{\text{obs}}(1+n_{\text{min}})}} = \frac{\varphi^2/5}{\varphi \cdot 10^6} = \frac{\varphi}{5 \times 10^6} = (1 + \sqrt{5}) \times 10^{-7} \in \mathbb{Z}[\sqrt{5}] \cdot 10^{-7}. \quad (37)$$

The suppression exponent $k_{\text{obs}}(1+n_{\text{min}}) = 2 \times 3 = 6$ combines $k_{\text{obs}} = 2$ and $n_{\text{min}} = 2$. The observer level $k_{\text{obs}} = 2$ is not a free choice. The cascade assigns each level a physical role:

$$\begin{aligned} k=0: & \quad n \gg 1, \alpha^{-1} && \text{projection pole (bulk zero-mode)} \\ k=1: & \quad n=11 && \text{electroweak attractor} \\ k=2: & \quad n=5 && \text{lock-in: the QED/atomic observable threshold} \\ k=3: & \quad n=3 && \text{molecular/hadronic boundary} \\ k=4: & \quad n=2 && \text{saturation mode.} \end{aligned} \quad (38)$$

Physical measurements of α are made at the lock-in level $k_{\text{obs}} = k = n_{\text{min}} = 2$, ($n_* = 5$): the unique level where $\xi^* \eta^* = \alpha^{-1}$ first stabilises. The physical observable that fixes this scale is the Rydberg constant $R_\infty = m_e c \alpha^2 / (2h)$: it is the atomic readout of the lock-in at $n_* = 5$, $k = 2$, giving the energy unit $\alpha \xi^* = m_e$ at the lock-in point. The QED/atomic threshold $k = 2$ is therefore not chosen by convention but identified by the structural condition that R_∞ is the boundary readout of the canonical pseudoinverse Π^+ at that level (derived in the companion Bohr–Rydberg paper; the gate is closed Level A in the QGT corpus). Only $k_{\text{obs}} = 2$ is used in this paper; the full cascade $k = 0, \dots, 4$ lies outside the scope of this paper.

The physical reading is this. The continuous hyperbolic flow $\dot{\xi} = -\Gamma_\tau \xi$, $\dot{\eta} = +\Gamma_\tau \eta$ preserves $\mathcal{H}_\tau = \xi \eta$ exactly. The discrete breathing map $\mathcal{B}_{k+1} = 2\mathcal{B}_k^{2/3}$ supplies the cycle structure sampled by low-energy measurements: one discrete breathing step converts coherent longitudinal amplitude into radiative outflow and is suppressed by the scale hierarchy $\mathcal{S}^6 = 10^6$, bringing the net effect to $(1 + \sqrt{5}) \times 10^{-7}$ per step. The confined mode $n_{\text{conf}} = 7$ contributes through the holonomy correction ε_7 separately; its role in f_{breath} is indirect, it is the non-tiling mode whose absence from the boundary forces the lock-in structure that determines ω_ξ and ω_η .

5.1.3 Self-consistent fixed point

$$\alpha_{\text{CODATA}}^{-1} = \frac{\alpha_{\text{QGT}}^{-1} - c_{\text{EW}} \varepsilon_7}{1 - f_{\text{breath}}}. \quad (39)$$

Table 1: Numerical chain from static pole to CODATA 2022. No CODATA value is inserted. The final residual 7×10^{-9} corresponds to 0.3σ of the CODATA 2022 uncertainty. The breathing back-action scale is derived in Appendix G from the rank-5 filter hierarchy and the boundary channel rates.

Step	α^{-1}	Residual from CODATA 2022
Static pole $(5 + 3\sqrt{5})^2$	137.082039	4.6×10^{-2}
$-c_{\text{EW}} \varepsilon_7$	137.035955	4.4×10^{-5}
$\div (1 - f_{\text{breath}})$	137.035999170	7×10^{-9}
CODATA 2022[4]	137.035999177(21)	(reference)

Remark 7 (Discrete robustness check). The numerical agreement is not obtained by tuning a continuous parameter. The cascade depends on three discrete structural choices: the confinement mode $n \in \mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$, the electroweak multiplicity $c \in \{5, 8, 13, 21\}$ (the only values with a structural derivation within the rank-5 filter), and the correction type (additive or multiplicative). This defines the complete space of structurally motivated alternatives; continuous deformations are excluded because the derivation contains no continuous free parameter. Natural counterfactual variants are tabulated immediately below. Only the structurally forced combination ($n = 7, c = 13$, multiplicative) reaches 7×10^{-9} ; every natural variant is worse by at least a factor of 40.

Table 2: Discrete sensitivity analysis. All structurally motivated alternatives to the selected combination ($n = 7, c = 13$, multiplicative) are listed with their residual from CODATA 2022. Rows varying n use the uniform residue $\varepsilon_n = \sqrt{5}\pi/n - 1$ of Lemma 3, all other factors held fixed: $\alpha^{-1}(n) = (\alpha_{\text{QGT}}^{-1} - c_{\text{EW}} \varepsilon_n)/(1 - f_{\text{breath}})$.

n	c	type	α^{-1}	error
7	13	mult. (this work)	137.035 999 170	7×10^{-9}
3	13	mult.	≈ 119.64	1.7×10^1
5	13	mult.	≈ 131.82	5.2
11	13	mult.	≈ 141.78	4.7
7	5	mult.	≈ 137.064	2.8×10^{-2}
7	8	mult.	≈ 137.053	1.7×10^{-2}
7	21	mult.	≈ 137.008	2.8×10^{-2}
7	13	additive	$\approx$ 137.035 999 846	3.3×10^{-7}

The result 137.035 999 170 differs from CODATA 2022 137.035 999 177(21) by 7 in the last digit: a discrepancy of 7×10^{-9} , which is 0.3σ of the experimental uncertainty [4]. The breathing back-action scale $f_{\text{breath}} = (1 + \sqrt{5}) \times 10^{-7}$ is derived from first principles in Appendix G: no CODATA value enters that derivation.

6 The gauge structure as a boundary residue

The pair $(r, |\mathcal{C}_{\text{obs}}|) = (5, 3)$ that fixes α^{-1} also organises the dimensionless gauge sector of the Standard Model. **Scope of this section.** The results below are the *gauge skeleton*: they determine the group structure $U(1) \times SU(2) \times SU(3)$, the tree-level mixing angle $\sin^2 \theta_W$, and the algebraic origin of the colour triplet at the level of representations. They do not determine physical masses, decay widths, running coupling constants, or any quantity that requires the full perturbative renormalisation group. These are internal *consistency checks* of the same projection that fixes α^{-1} , not independent derivations of the full Standard Model dynamics. A QFT reader should read every result in this section as: “the rank-5 projection forces this group structure at tree level; QED/QCD dynamics on top of that structure remain the correct tool for all quantitative predictions.”

6.1 The electroweak mixing angle: tree-level value and structural closing

The root matrix $\mathbf{M}$ of Section 4 has diagonal entries $(r, r) = (5, 5)$ and off-diagonal entries $(\kappa^2 |\mathcal{C}_{\text{obs}}|, |\mathcal{C}_{\text{obs}}|) = (15, 3)$. The gauge tensor on the two-dimensional dynamical sector is built from the

scalar contractions of $\mathbf{M}$: the diagonal entry $r = 5$ weights the longitudinal channel and the off-diagonal entry $|\mathcal{C}_{\text{obs}}|/2 = 3/2$ weights the transverse channel, giving:

$$\mathbf{G} = \beta \begin{pmatrix} r & 0 \\ 0 & |\mathcal{C}_{\text{obs}}|/2 \end{pmatrix} = \beta \begin{pmatrix} 5 & 0 \\ 0 & 3/2 \end{pmatrix}, \quad (40)$$

giving:

$$\sin^2 \theta_W = \frac{G_{22}}{\text{tr} \mathbf{G}} = \frac{3}{13} \approx 0.2308. \quad (41)$$

The measured value at the Z-pole is 0.23122 ± 0.00003 [5]: a gap of 0.19% (0.75σ), consistent with the tree-level structural value prior to radiative running. The pair $(r, |\mathcal{C}_{\text{obs}}|) = (5, 3)$ fixes both α^{-1} and $\sin^2 \theta_W$ as two numerical outputs of the same $\mathbf{M}$; they are not independent results. This is a tree-level boundary value; radiative corrections are not computed here. The same structural objects that close α^{-1} to CODATA also close $\sin^2 \theta_W$, with no new parameter. Define the geometric entropy suppression factor:

$$\Upsilon_{\text{geom}} = \frac{\alpha^{-1}}{\mathcal{S}^2} = \frac{\varphi^4}{r} \in \mathbb{Z}[\sqrt{5}], \quad (42)$$

where $\mathcal{S} = n_{\text{min}} \cdot r = 10$ is the spectral norm of the rank-5 filter (already derived in Section 4). The ξ -channel (electroweak) correction uses the same holonomy residue ε_7 and breathing factor f_{breath} as the η -channel correction for α^{-1} (the holonomy and breathing corrections of Section 5), but suppressed by Υ_{geom} :

$$\boxed{\sin^2 \theta_W^{\text{closed}} = \frac{\sin^2 \theta_W^{\text{QGT}} \left(1 + \Upsilon_{\text{geom}} \cdot \frac{|\mathcal{C}_{\text{obs}}|}{n_{\text{conf}}} \cdot \varepsilon_{n_{\text{conf}}} \right)}{1 - \Upsilon_{\text{geom}} \cdot f_{\text{breath}}}} \quad (43)$$

Substituting $\sin^2 \theta_W^{\text{QGT}} = 3/13$, $\Upsilon_{\text{geom}} = \varphi^4/r$, $|\mathcal{C}_{\text{obs}}|/n_{\text{conf}} = 3/7$, $\varepsilon_7 = \sqrt{5}\pi/7 - 1$, $f_{\text{breath}} = (1 + \sqrt{5}) \times 10^{-7}$:

$$\sin^2 \theta_W^{\text{closed}} \approx 0.23125, \quad (44)$$

within 0.75σ of the Z-pole electroweak value 0.23122 ± 0.00004 [5]. No parameter is introduced: Υ_{geom} , ε_7 , and f_{breath} are all forced by the rank-5 projection structure already determined in deriving α^{-1} . The same denominator $c_{\text{EW}} = 2r + |\mathcal{C}_{\text{obs}}| = 13$ that appears in $\sin^2 \theta_W = 3/13$ governs the holonomy correction to α^{-1} : the electroweak structure and the fine-structure constant are two scalar contractions of the same Fibonacci matrix $\mathbf{M}$ [6].

6.2 Electroweak sector from the two chiral channels

The two chiral channels give the boundary coupling tensor $\alpha = (\alpha^{(G)}/\sqrt{\mathcal{K}_\tau}) \text{diag}(\xi, \eta)$, with anisotropy $\xi^*/\eta^* = n_{\text{sat}}/n_* = 11/5$ at the lock-in (the ratio of the saturation mode $n_{\text{sat}} = 11 = \max \mathcal{N}_{\text{adm}}$ to the lock-in mode $n_* = 5$, both forced by $\mathcal{N}_{\text{adm}}$). Together they carry the $U(1) \times \text{SU}(2)$ electroweak sector. Majorana self-conjugacy protects custodial symmetry $M_{W^+} = M_{W^-}$ ($\rho_{\text{sym}} = 1$) as the palindromic symmetry of the mode-weight tensor (Appendix E).

6.3 Colour from the confined mode

The nontrivial structural step is not the Lie closure itself (any triplet $\mathbb{C}^3$ closes to $\mathfrak{su}(3)_c$ as traceless anti-Hermitian endomorphisms) but the structural necessity of the confined triplet $K_7 \subset \ker \Pi$. Mode $n_{\text{conf}} = 7$ is the unique element of $\mathcal{N}_{\text{adm}}$ that does not factorise within the Fibonacci basis $\{2, 3, 5\}$: it cannot appear at the boundary as a free mode and is therefore *forced* into $\ker \Pi$. Once confined, $K_7 \cong \mathbb{C}^3$ is the minimal realisation of a triplet in $\ker \Pi$, and its three pairwise $\text{SU}(2)$ subalgebras close to $\mathfrak{su}(3)_c$ (proof below). This is the minimal colour gauge skeleton; full QCD dynamics are not derived here.

Table 3: Standard Model gauge-sector skeleton organised by $\mathbf{M}$ and the boundary/kernel structure of Π . Asterisk: pre-dimensional weights only; physical masses require further steps (open).

Object	Origin	Physical role
$\mathbf{M}$ (mult. by $5+3\sqrt{5}$ in $\mathbb{Z}[\sqrt{5}]$)	Fibonacci fixed point in $\mathbb{Z}[\sqrt{5}]$	Root matrix
$\alpha^{-1} = \ \mathbf{M}^2\ _{\mathbb{Z}[\sqrt{5}]}$	Squared norm of $\mathbf{M}$	EM operating point
$\sin^2 \theta_W = 3/13$	Ratio $G_{22}/\text{tr} \mathbf{G}$	EW mixing (check)
$U(1) \times \text{SU}(2)$	Two-channel form α	EW sector
$\rho_{\text{sym}} = 1$	Palindrome of D_{eff}	Custodial symmetry
$\mathfrak{su}(3)_c$	Lie closure on $K_7 \subset \ker \Pi$	Colour skeleton

6.4 Summary

7 Consequences and open problems

This section separates one α -sector prediction from two further directions. The prediction concerns physics not yet observed; the further directions indicate open extensions.

Prediction 1 — The time quantum $\delta\phi = \pi\alpha$. The minimum observable phase step on the projected boundary is:

$$\delta\phi = \pi\alpha = \frac{\pi}{\alpha^{-1}} \approx 2.296 \times 10^{-2} \text{ rad.} \quad (45)$$

This is not a numerical coincidence. It is the elementary half-cycle of the temporal fibre $S^1(\tau)$ at the admissible mode $n = \alpha^{-1} = 137$: the number of elementary phase steps $\pi\alpha$ that fit in one half-cycle π of the fibre is exactly $\lfloor \alpha^{-1} \rfloor = 137$. The time quantum $\delta\phi = \pi\alpha$ is the boundary analogue of $\hbar$: the granularity of observable temporal evolution, expressed in phase units. It sets the resolution floor of the boundary clock and determines the scale at which quantum mechanics operates, which is why $\alpha^{-1} = 137$ and the fine-structure constant are the same structural object read in different units.

Falsification: any experiment that resolves temporal phase below $\delta\phi = \pi\alpha$ may probe the projection-induced discretisation of the boundary clock. A boundary-clock observable of this construction should exhibit a projection-induced phase floor at $\delta\phi \sim \pi\alpha$, structurally distinct from thermal or shot noise.

Further direction 1 — Chiral modes and gravitational sector. The scalar α^{-1} is the invariant norm of the boundary operating point. On ∂V , however, the projection has two chiral channels (ξ, η) that cannot be identified because Π is non-invertible. The scalar pole therefore acquires a two-channel tensorial form:

$$\alpha = \frac{\alpha^{(G)}}{\sqrt{\mathcal{K}_\tau}} \begin{pmatrix} \xi & 0 \\ 0 & \eta \end{pmatrix}, \quad \mathcal{K}_\tau = \xi\eta = \alpha^{-1}. \quad (46)$$

The longitudinal component $\alpha_{\parallel} \propto \xi$ is the curvature/gravitational readout; the transverse component $\alpha_{\perp} \propto \eta$ is the radiative/electromagnetic readout. At lock-in, $\alpha_{\parallel}/\alpha_{\perp} = \xi^*/\eta^* = 11/5$. A QED experiment probes $\alpha_{\perp}$; the scalar α^{-1} is the $\mathbb{Z}[\sqrt{5}]$ -norm of the full boundary object α . The identification with the gravitational sector is a structural readout of the same projection; the full dynamical derivation requires derivation of R_5 from the projection structure, an open problem. The same non-invertibility of Π produces an unconditional irreversibility: each projection cycle loses a fraction

$$\delta_U = 1 - \frac{3\pi}{(2\sqrt{5})^{3/2}} \neq 0 \quad (47)$$

of bulk norm to the kernel. The inequality $\delta_U \neq 0$ is a theorem of pure mathematics (Lindemann–Weierstrass: $3\pi = (2\sqrt{5})^{3/2}$ would require $\pi \in \overline{\mathbb{Q}}(\sqrt{5})$, which is false), so the arrow of time is a structural output of $\ker \Pi \neq \{0\}$, not an additional postulate. The gauge structure of the rank-5 filter and the kernel geometry are developed further in [6, 7].

Further direction 2 — The cosmic microwave background from rank-5 projection. As an outlook beyond the α -sector, the same rank-5 projection makes structural predictions for the CMB power spectrum [8]. The key structural outputs are:

$$\Omega_b h^2 = \frac{\sqrt{5}}{100} \approx 0.02236 \quad (\text{palindromic Silk residue, Prop. 4}), \quad (48)$$

$$\Omega_c h^2 = \frac{1}{5\phi} \approx 0.12361 \quad (\text{structural theorem of } \ker \Pi, \text{ Props. 5–6}), \quad (49)$$

$$\tau_{\text{reion}} = \frac{7}{130}, \quad H_0 \simeq 67.9 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (50)$$

Numerically: six TT peak amplitudes reproduced at RMS 5.20% with zero free cosmological parameters; six TE nodes at RMS 1.7%; passed to CAMB with all six structural inputs, $\chi^2_V = 1.18$ against Planck 2018, compared to Λ CDM ($\chi^2_V = 1.04$) with six fitted parameters. The residual $\chi^2_V = 1.18$ is the structural distance from the Planck maximum-likelihood point with zero degrees of freedom conceded.

The dark matter density $\Omega_c h^2 = 1/(5\phi)$ follows from the Fibonacci fixed point at $n_* = 5$; $\tau_{\text{reion}} = 7/130$ follows from the confined mode $n_{\text{conf}} = 7 \in \ker \Pi$; H_0 follows from the analytic-return suppression of the breathing equation. These are not fitted: they are outputs of the same projection that fixes α^{-1} and $\sin^2 \theta_W$. The scalar amplitude A_s is connected to the Heegner kernel at the CM point τ_{163} ; its detailed derivation is in [8].

7.1 The α -sector field equation set

The closed α -sector of the construction is now collected. We collect all equations in a single self-contained set, with operators defined precisely, as a reference for the reader.

α -sector field equations, no free parameters in the gauge/EM sector.

Spacetime and fundamental field:

$$\mathcal{M}^5 = \mathbb{R}^4 \times S^1(\tau), \quad X^A = (x^\mu, \tau).$$

$$\Xi = C\bar{\Xi} \quad (\text{Majorana}), \quad \mathcal{D}_5 \Xi = 0, \quad \mathcal{D}_5 := \Gamma^A \nabla_A.$$

(Operators: $\mathcal{D}_5 =$ bulk Dirac; $D_\mu =$ boundary covariant derivative; $D = x\partial_x =$ Pincherle scale operator. These are distinct.)

Conservation law, the single structural axiom:

$$J^A[\Xi] = \bar{\Xi} \Gamma^A \Xi, \quad \nabla_A J^A = 0.$$

Projection and analytic signal decomposition:

$$\Pi : \mathcal{H}_{\text{bulk}} \rightarrow \mathcal{H}_{\text{bdy}}, \quad \ker \Pi \neq \{0\}, \quad \text{rank } \Pi = 5.$$

$$\Psi = \Pi \Xi = \text{Re}(\Xi), \quad \Xi_{\text{ker}} = -i \mathcal{H}[\Psi], \quad \Xi = \Psi + i \Xi_{\text{ker}}.$$

Projected Noether balance (boundary form of $\nabla_A J^A = 0$):

$$\partial_\xi J^\xi = -\nabla_\eta J^\eta.$$

Boundary field equation, unified nonlinear Dirac:

$$(\gamma^\mu D_\mu + \alpha \xi + i\alpha \eta \gamma^5) \Psi = \frac{\delta \mathcal{S}_{\text{ker}}^{(5)}[\Psi]}{\delta \bar{\Psi}}, \quad \mathcal{S}_{\text{ker}}^{(5)}[\Psi] = -\text{Tr}_{\mathcal{H}_{\text{ret}}^{(5)}}(\rho_\Psi \log \rho_\Psi).$$

Channel dynamics, hyperbolic rotation on Σ_α : $\xi \eta = \alpha^{-1}$:

$$\dot{\xi} = -\Gamma_\tau \xi, \quad \dot{\eta} = +\Gamma_\tau \eta, \quad \xi \eta = \alpha^{-1}, \quad \Gamma_\tau = \frac{\varphi}{\sqrt{5}}.$$

$$\theta_{\log} := \frac{1}{2} \log(\eta/\xi), \quad \dot{\theta}_{\log} = \Gamma_\tau \text{ (physical time rate)}.$$

Scale operator and palindromic closure:

$$D = x \partial_x, \quad D_{\text{eff}} = \Pi D \Pi^*, \quad D_{\text{eff}}^2 - k D_{\text{eff}} + 1 = 0.$$

Boundary filter, spectral response of the rank-5 code subspace:

$$H_\alpha(z) = \frac{P_4(z)}{1 - \alpha z}, \quad P_4(z) = \sum_{m=0}^4 (\log \varphi)^m z^m, \quad z_{\text{pole}} = \alpha^{-1} = (5 + 3\sqrt{5})^2.$$

Breathing amplitude, discrete boundary recursion:

$$B[\Psi] = \|\Psi\|_{\text{bdy}}^2, \quad B_{n+1} = 2B_n^{2/3}, \quad B_* = 8.$$

The logical chain is:

$$\Pi \Rightarrow \mathcal{C}_{\text{bdy}}^{(5)} \Rightarrow H_\alpha \Rightarrow \alpha^{-1}.$$

The classical field theories are chiral limits of the boundary equation:

$$\begin{aligned} \eta \rightarrow 0: \quad (\gamma^\mu D_\mu + \alpha \xi) \Psi &= 0 & \longrightarrow \text{GR (Einstein equations),} \\ \xi \rightarrow 0: \quad (\gamma^\mu D_\mu + i\alpha \eta \gamma^5) \Psi &= 0 & \longrightarrow \text{EM (Maxwell equations).} \end{aligned}$$

The nonlinear right-hand side $\delta \mathcal{S}_{\text{ker}}^{(5)} / \delta \bar{\Psi}$ is what Maxwell and Einstein do not have: the back-action of the Hilbert quadrature $\Xi_{\text{ker}} = -i\mathcal{H}[\Psi]$ on the boundary dynamics. It generates mass, stabilises the operating point, and makes α^{-1} a *derived* pole of H_α rather than a free parameter.

Framework	Central equation	Input	Output
Maxwell	$\partial_\nu F^{\mu\nu} = J^\mu$	fields + charges	EM given the source
Einstein	$G_{\mu\nu} = 8\pi G T_{\mu\nu}$	metric + matter	geometry given content
Dirac	$(\gamma^\mu \partial_\mu - m) \psi = 0$	mass m	fermions in vacuum
Thermodynamics	$dS \geq 0$	equilibrium state	irreversibility
This work	$(\gamma^\mu D_\mu + \alpha \xi + i\alpha \eta \gamma^5) \Psi = \delta \mathcal{S}_{\text{ker}}^{(5)} / \delta \bar{\Psi}$	$\nabla_A J^A = 0$ only	α^{-1} , $\sin^2 \theta_W$, EM, GR as outputs

In summary, a bulk Majorana spinor projected onto a rank-5 boundary loses its Hilbert quadrature to the kernel, and that quadrature returns as the nonlinear term that stabilises the boundary pole at $\alpha^{-1} = 137.036$.

Beyond the closed α -sector and the structural electroweak matching of Section 6, the remaining open extensions are the full fermionic transporter beyond the charged-lepton sector [9], QCD dynamics, the

full scheme-dependent electroweak renormalisation-group evolution $\sin^2 \hat{\theta}_W(Q^2)$, and the gravitational sector.

Sommerfeld’s constant is not a free parameter of nature. It is what a Majorana bulk cannot deliver across its own projection interface. The bulk carries a coherent phase around $S^1(\tau)$; the projection to the observable boundary discards part of that coherence into a physical kernel. What the kernel holds back exerts a return tension on the boundary dynamics. That tension, stabilised by the Fibonacci structure of the rank-5 filter, fixes the pole at $\alpha^{-1} = 137.036$. The number is not mysterious. It is the unique stable operating point of the only autosimilar transport law compatible with a Majorana spinor on a compact fibre under a non-invertible projection. The cited extensions to the lepton sector and cosmology develop the same projection principle beyond the α -sector [10].

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Conflict of interest

The author declares no conflicts of interest.

Data availability

A standalone verification script (`alpha_check.py`; 30-digit arithmetic) asserting every exact identity, the full cascade against CODATA 2022, and each counterfactual row of Table 2 accompanies the manuscript. No data were generated or analysed for this study.

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A The observability obstruction: formal statement

The following gives the precise statement of the non-derivability result used in Section 2. The argument shows that within any $\mathbb{R}^4$ -based framework, no internal mechanism can produce the admissibility structure that a derivation of α would require.

Setting. Let $\mathcal{A}$ be the C^* -algebra of observables of a field theory formulated intrinsically on $\mathbb{R}^4$ (Standard-Model or General-Relativity type). An *admissibility projection* is a map $E : \mathcal{A} \rightarrow \mathcal{A}_{\text{adm}}$ satisfying simultaneously:

- (i) $\mathcal{A}_{\text{adm}} \subset \mathcal{A}$ is a subalgebra;
- (ii) $E \circ E = E$ (idempotent);
- (iii) $\ker E \neq \{0\}$ (non-trivial kernel);
- (iv) $\ker E$ contains degrees of freedom physically eliminated prior to observation, not merely gauge-redundant.

Non-derivability result. No admissibility projection satisfying (i)–(iv) can be derived from the internal structure of an $\mathbb{R}^4$ -based field theory.

Proof. Gauge fixing eliminates descriptive redundancy, not physical degrees of freedom: condition (iv) fails. Coarse-graining produces practical inaccessibility, not principled inadmissibility: in the idealised limit of complete experimental control the eliminated degrees of freedom become accessible, so (iv) fails. Decoherence is a dynamical process depending on an external environment and coupling: it is not a fixed idempotent map on $\mathcal{A}$, so (ii) fails. Renormalisation reorganises the theory across scales without identifying degrees of freedom inadmissible at all scales: (iv) fails. Algebraic QFT supplies observable algebras through additional axioms (locality, Haag duality): these are supplementary structural postulates, not consequences of the field dynamics, so they are not internal to the $\mathbb{R}^4$ theory.

The only remaining possibility is an intrinsic automorphic selection. But every $*$ -automorphism $\phi : \mathcal{A} \rightarrow \mathcal{A}$ of a C^* -algebra is isometric: the C^* -identity $\|a^*a\| = \|a\|^2$ is preserved under ϕ , giving $\|\phi(a)\| = \|a\|$ for all a , hence $\ker \phi = \{0\}$ [2]. Condition (iii) is therefore violated. No internal mechanism of the $\mathbb{R}^4$ -theory satisfies (i)–(iv). $\square$

Corollary (non-derivability of α in $\mathbb{R}^4$). A dimensionless constant is derivable as a structural invariant of $\mathcal{A}$ only if the theory internally specifies the admissible subalgebra in which that invariant is evaluated. By the result above, no such subalgebra is specifiable from within $\mathbb{R}^4$. Therefore no dimensionless coupling constant, in particular α , can be derived from within an $\mathbb{R}^4$ -based theory. $\square$

B Uniqueness of $H_\alpha(z)$ under the QGT constraints

We prove that $H_\alpha(z) = P_4(z)/(1 - \alpha z)$ is the unique rational transfer function of D_{eff} satisfying the following four QGT constraints: (i) rank $r = 5$; (ii) the palindromic operator equation (19); (iii) a single spectral pole at $z_{\text{pole}} = \alpha^{-1}$; (iv) the Noether normalisation $\mathcal{K}_\tau = \xi\eta = \alpha^{-1}$. Uniqueness is conditional on these constraints; it does not exclude rational realisations built on different operator or basis choices.

The proof has three independent parts: uniqueness of the function class, uniqueness of the denominator, and uniqueness of the pole value.

Part I: Uniqueness of the function class (numerator).

The Pincherle dilatation $D = \tau \partial_\tau$ acts multiplicatively on the autosimilar hierarchy $\{\varphi^k\}_{k \geq 0}$ of $S^1(\tau)$. Its eigenvalue equation is $D \cdot \varphi^k = k\varphi^k$, so the eigenvalues are the non-negative integers and the eigenbasis is $\{x^k\}_{k=0}^\infty$ in the Mellin variable. Under projection to the boundary, the continuous Mellin basis is compressed to the finite set retained by the truncated SVD of rank $r = 5$.

The canonical Mellin-diagonal basis adapted simultaneously to (a) the multiplicative scaling φ^k and (b) the boundary metric of the rank-5 projection is the *log-cumulant* basis $\{v_m : v_m(x) = (\log x)^m, m = 0, \dots, 4\}$. This follows because $(\log x)^m$ are the *cumulants* of the log-normal distribution associated to the Binet form $F_n = (\varphi^n - \psi^n)/\sqrt{5}$: they diagonalise the action of D_{eff} on the rank-5 image.

Any other basis would introduce off-diagonal terms in the action of D_{eff} : in particular, a polynomial basis $\{1, x, x^2, \dots\}$ does not diagonalise D_{eff} because $D_{\text{eff}}(x^k) = kx^k + \text{lower}$ in Mellin space, generating coupling between levels. The log-cumulant basis is therefore the canonical Mellin-diagonal basis satisfying (i) and (ii) simultaneously; any other basis introduces off-diagonal coupling in D_{eff} .

With this basis fixed, the numerator polynomial is:

$$P_4(z) = \sum_{m=0}^4 (\log \varphi)^m z^m, \quad (\text{B.51})$$

of degree $r - 1 = 4$ (one degree consumed by the single denominator pole). No free parameter enters: the coefficients $(\log \varphi)^m$ are fixed by the golden eigenvalue φ of D_{eff} .

Note on independence. The value of α^{-1} depends exclusively on the denominator pole, not on the normalisation of P_4 . The numerator fixes the finite-rank response shape (which modes are transmitted and with what weight); the denominator fixes the unique boundary operating point. Any rescaling of P_4 leaves the pole $z_{\text{pole}} = \alpha^{-1}$ unchanged.

Part II: Uniqueness of the denominator.

The palindromic operator equation (19), $D_{\text{eff}}^2 - kD_{\text{eff}} + 1 = 0$, implies that D_{eff} and D_{eff}^{-1} satisfy the same characteristic polynomial. The transfer function $H(z)$ must therefore be *self-reciprocal*: $H(z) = z^{\deg H} \cdot H(1/z)/H(1/z)|_{z=\infty}$ up to a scalar. This forces the denominator to be palindromic.

We classify all palindromic denominators compatible with condition (iii) (single pole) and (iv) (single Noether invariant $\mathcal{K}_\tau$):

- *Degree $d = 0$* : no denominator, no pole. Excluded by (iii).
- *Degree $d = 1$* : denominator $(1 - \alpha z)$, single pole at $z_{\text{pole}} = \alpha^{-1}$. Compatible with (iii) and (iv). **Admitted.**
- *Degree $d \geq 2$, palindromic*: by Vieta's formulas, the product of all poles equals the ratio of constant to leading coefficient, which for a palindromic polynomial of degree d satisfies $\prod_{i=1}^d z_{\text{pole}}^{(i)} = \pm 1$. A palindromic denominator of degree $d = 2$ has poles z_1, z_2 with $z_1 z_2 = 1$, i.e. $z_2 = 1/z_1$. These two poles produce two independent Noether invariants: one at z_1 and one at $z_2 = 1/z_1$. But condition (iv) requires a *single* invariant $\mathcal{K}_\tau = \xi \eta = \alpha^{-1}$. Two poles over-constrain the system and are incompatible with a single boundary invariant. Excluded by (iv) for all $d \geq 2$.

The unique admissible denominator is therefore $(1 - \alpha z)$, giving $H_\alpha(z) = P_4(z)/(1 - \alpha z)$.

Part III: Uniqueness of the pole value.

The pole $z_{\text{pole}} = \alpha^{-1}$ must be the dominant eigenvalue of D_{eff} on the golden lattice. By the spectral curvature $\kappa = \sqrt{5}$ and the Fibonacci fixed point $F_{n_*} = n_* = r = 5$ (eq. (27)), the dominant eigenvalue is:

$$z_{\text{pole}} = (F_5 + F_4\sqrt{5})^2 = (5 + 3\sqrt{5})^2 = 70 + 30\sqrt{5}. \quad (\text{B.52})$$

The two Fibonacci coordinates $(F_5, F_4) = (5, 3)$ are fixed by the lock-in without freedom. The conjugate lattice element is $x_- = 5 - 3\sqrt{5} \approx -1.708$, with $(x_-)^2 = 70 - 30\sqrt{5} \approx 2.918$. As a z -value the conjugate pole would sit at $z_- = (x_-)^{-2} = 1/(70 - 30\sqrt{5}) \approx 0.343$, which lies *inside* the unit disk. It therefore generates no pole in H_α : it is absorbed into the zeros of the MA numerator $P_4(z)$, not into the denominator. The unique dominant pole is $z_{\text{pole}} = 70 + 30\sqrt{5} = \alpha_{\text{QGT}}^{-1}$.

Parts I–III together show that $H_\alpha(z) = P_4(z)/(1 - \alpha z)$ is the unique rational transfer function satisfying conditions (i)–(iv). $\square$

C The boundary code subspace: Mellin-diagonal closure

This appendix supplies the spectral argument that the linearisation $\mathcal{L}_*$ of the nonlinear boundary Dirac equation has exactly five observable eigenmodes, closing the logical gap between the rank-5 projection and the code subspace claim of Section 3.8.

The exponent $2/3$ from dimensional reduction. The return tension has the form $V'(|\Psi|) \propto |\Psi|^{2/3}$. This exponent is not a modelling choice: it follows from the unique dimensional relationship between bulk volume and boundary surface. Let $[\mathcal{B}] \sim L^3$ be the bulk coherence density (3D) and $[\sigma] \sim L^2$ the boundary amplitude (2D). The power law $\sigma = \mathcal{B}^\beta$ with dimensional consistency $L^2 = (L^3)^\beta$ gives $\beta = 2/3$ uniquely. This is precisely the exponent of the breathing equation $\mathcal{B}_{k+1} = 2\mathcal{B}_k^{2/3}$ of Section 3.12: the same $3D \rightarrow 2D$ projection that drives the discrete Noether balance determines the return tension at the fixed point. The gradient of the effective potential is therefore:

$$V'(|\Psi|) = \frac{\delta \mathcal{S}_{\text{ker}}}{\delta |\Psi|} = \beta_* |\Psi|^{2/3}, \quad \beta_* = -\frac{2}{3} \lambda_{\text{eff}}, \quad \lambda_{\text{eff}} = \alpha^2 \mathcal{K}_\tau = \alpha, \quad (\text{C.53})$$

where $\lambda_{\text{eff}} = \alpha$ at the lock-in fixed point $\xi \eta = \mathcal{K}_\tau = \alpha^{-1}$. The Thomas–Fermi potential $V_{\text{eff}}(|\Psi|) = \frac{3}{5} \beta_* |\Psi|^{5/3}$ is the energy functional corresponding to the same $3D \rightarrow 2D$ reduction. No free parameter enters: both the exponent and the coefficient are fixed by the projection structure.

Lemma (Mellin diagonality of $\mathcal{L}_*$). *On the scale-covariant sector at the fixed point Ψ_* , the linearised tangent operator $\mathcal{L}_*$ is diagonal in the Mellin domain: it acts as a scalar multiplication on each Mellin mode, with no inter-level mixing.*

Proof. The bulk is multiplicative: the natural variable is $x = e^\tau$ and the Pincherle dilatation $\mathcal{D} = x\partial_x = \partial_{\log x}$ is the scale generator. In the logarithmic variable, $\gamma^\mu D_\mu$ acts as a Mellin multiplier (multiplication by the Mellin variable t). At the fixed point Ψ_* , the field is scale-covariant: $\Psi_*(ax) = a^s \Psi_*(x)$ for some s (the fixed point belongs to the scale-invariant sector of the autosimilar boundary). For a pure power-law nonlinearity $|\Psi|^s \Psi$ evaluated at a scale-covariant Ψ_* , the Mellin transform acts as a shift $t \mapsto t + (s+1)/2$, not a convolution: the nonlinearity multiplies each Mellin mode by a scalar, not by a mode-coupling kernel. This scale-covariance at Ψ_* is the key: it is what makes the linearisation Mellin-diagonal rather than a general mode-mixing operator. The Fréchet derivative of $|\Psi_*|^{2/3} \Psi$ around Ψ_* contains $|\Psi_*|^{2/3} \chi$ and $|\Psi_*|^{-1/3} \text{Re}\langle \Psi_*, \chi \rangle \hat{\Psi}_*$: both are scalar multiplications on χ in the Mellin domain, not couplings between different Mellin levels. Therefore $\mathcal{L}_*$ is Mellin-diagonal on this sector. $\square$

Corollary ($\dim \text{Spec}_{\text{obs}} = 5$). A Mellin-diagonal operator preserves any span of Mellin modes. The rank-5 projection retains exactly the first five Mellin modes $v_m(x) \propto (\log x)^m$, $m = 0, \dots, 4$ (the log-cumulant basis of Section 4). Since $\mathcal{L}_*$ is Mellin-diagonal, it maps this span to itself: $\mathcal{L}_*(v_m) \in \text{span}\{v_0, \dots, v_4\}$ for each m . The observable spectrum is the restriction to this five-dimensional span: five eigenvalues $\lambda_0, \dots, \lambda_4$, one per retained Mellin mode. These are the eigenstates $v_0, \dots, v_4$ of the boundary code subspace. $\square$

D Structural consistency readings of the static pole

The static pole $\alpha_{\text{QGT}}^{-1} = 70 + 30\sqrt{5}$ admits five structural consistency readings, each arising from a different geometric position on the observable boundary region. These are equivalent readouts of the same projection pole, not independent derivations.

Formula for α^{-1}	Status	Position
$(5 + 3\sqrt{5})^2 = 70 + 30\sqrt{5}$	Exact in $\mathbb{Z}[\sqrt{5}]$	Noether surface Σ
$20\varphi^4$	Exact in $\mathbb{Z}[\sqrt{5}][\varphi]$	Σ
$n_{\min}^2 r \varphi^4$	Exact	Σ
$500\Gamma_\tau^4$	Exact	Domain wall DW_1
$\approx (e_E \pi)^{-2} \times 10^4$	Near-identity (diagnostic only)	Saturation wall

Here e_E is Euler's number; the near-identity is not in $\mathbb{Z}[\sqrt{5}][\pi]$ and is not used in any derivation.

E Electroweak modal-weight tensor

Rank- r non-invertibility forces the decomposition of $\rho_{\text{pre}} = r\beta$ into three boundary residues with coefficients $\{r, |\mathcal{C}_{\text{obs}}|, n_{\min}\}/2 = \{5, 3, 2\}/2$. The modal-weight tensor is:

$$\mathbf{T}_{\text{EW}} = \frac{\beta}{2} \begin{pmatrix} 7 & 3 \\ 3 & 3 \end{pmatrix}. \quad (\text{E.54})$$

The modal-weight tensor $\mathbf{T}_{\text{EW}}$ has chiral contractions:

$$\omega_- = \frac{\beta}{2}(7+3) = 5\beta, \quad \omega_+ = \frac{\beta}{2}(7-3) = 2\beta, \quad \omega_0 = \frac{3\beta}{2}. \quad (\text{E.55})$$

These are pre-dimensional boundary contractions of $\mathbf{T}_{\text{EW}}$, *not* its eigenvalues. The actual eigenvalues of $\mathbf{T}_{\text{EW}}$ are $\lambda_{\pm} = \frac{\beta}{2}(5 \pm \sqrt{13})$, which should not be confused with the chiral contractions $\omega_{\pm}$. The off-diagonal entry $3 = |\mathcal{C}_{\text{obs}}|$ encodes Majorana self-conjugacy. The custodial statement $\rho_{\text{sym}} = 1$ is the palindromic symmetry of the rank-5 boundary algebra; it protects the physical equality $M_{W^+} = M_{W^-}$ after the dimensional cascade. It does not assert equality of the pre-dimensional contractions ω_- and ω_+ . The dimensional cascade $E_k = \hbar\omega_0/(\alpha\mathcal{S}^k)$, with $\mathcal{S} = n_{\min} \cdot r = 10$, follows from the rank-5 filter structure; its full development lies outside the scope of this paper. Converting the pre-dimensional contractions $(\omega_-, \omega_+, \omega_0)$ into physical masses further requires the non-linear boundary closure, which remains open here.

F Colour algebra from the confined triplet

Proposition 4. *The Lie closure of the three pairwise $\text{SU}(2)$ subalgebras on $K_7 \cong \mathbb{C}^3$ is $\mathfrak{su}(3)_c$.*

Proof. Let $\{\lambda_1, \dots, \lambda_8\}$ be the Gell-Mann matrices. Define $h_{13} = (\lambda_3 + \sqrt{3}\lambda_8)/2$ and $h_{23} = (-\lambda_3 + \sqrt{3}\lambda_8)/2$. The subalgebras $\mathfrak{s}_{12} = \text{span}\{\lambda_1, \lambda_2, \lambda_3\}$, $\mathfrak{s}_{13} = \text{span}\{\lambda_4, \lambda_5, h_{13}\}$, $\mathfrak{s}_{23} = \text{span}\{\lambda_6, \lambda_7, h_{23}\}$ nominally contribute 9 generators. The Cartan elements overlap: $h_{13} + h_{23} = \sqrt{3}\lambda_8$ and $h_{13} - h_{23} = \lambda_3$, reducing the span to exactly $\{\lambda_1, \dots, \lambda_8\}$. Key commutators: $[\lambda_1, \lambda_2] = 2i\lambda_3$, $[\lambda_4, \lambda_5] = 2ih_{13}$, $[\lambda_6, \lambda_7] = 2ih_{23}$, $[\lambda_1, \lambda_4] = i\lambda_7$, all closing within $\text{span}\{\lambda_1, \dots, \lambda_8\}$. Hence $\dim = 8 = \dim \mathfrak{su}(3)_c$. $\square$

This is the minimal colour gauge skeleton from $K_7 \cong \mathbb{C}^3$; the full dynamical content of QCD is not derived.

G First-principles derivation of f_{breath}

This appendix gives the derivation of the breathing back-action scale 10^{-7} from the rank-5 filter hierarchy and the boundary channel rates, without free parameters.

Two independent inputs.

Input 1 (filter structure, exact). The observer level $k_{\text{obs}} = n_{\text{min}} = 2$: the lowest non-trivial boundary readout requires the minimal spinorial pairing (two-fold covering $\text{SU}(2) \rightarrow \text{SO}(3)$, fixed by $n_{\text{min}} = 2$). The filter renormalisation depth is $1 + n_{\text{min}} = 3$: one bulk return step plus the $n_{\text{min}} = 2$ minimal boundary spinorial distinctions. Hence the suppression exponent is:

$$k_{\text{obs}}(1 + n_{\text{min}}) = 2 \times 3 = 6. \quad (\text{G.56})$$

The boundary ARMA filter has scale decade $\mathcal{S} = n_{\text{min}} \cdot r = 2 \times 5 = 10$. The total scale suppression at observer level k_{obs} is therefore:

$$\mathcal{S}^{k_{\text{obs}}(1+n_{\text{min}})} = 10^{2 \times 3} = 10^6. \quad (\text{G.57})$$

Every factor in (G.57) is determined by the rank-5 filter structure: $n_{\text{min}} = 2$, $r = 5$, $k_{\text{obs}} = 2$.

Input 2 (boundary channel rates, exact). From the lock-in anisotropy $\xi^*/\eta^* = n_{\text{sat}}/n_* = 11/5$ (Section 3.8), the two channels present to the boundary observer at rates:

$$\omega_\xi = \Gamma_\tau \cdot \frac{n_*}{n_{\text{sat}}}, \quad \omega_\eta = \Gamma_\tau \cdot \frac{n_{\text{sat}}}{n_*}, \quad \omega_\xi \cdot \omega_\eta = \Gamma_\tau^2 = \frac{\varphi^2}{5}. \quad (\text{G.58})$$

The derivation. The back-action per discrete breathing step on the observed α^{-1} is the product (G.58), suppressed by (G.57) and renormalised by the Majorana coupling $\varphi = \kappa \Gamma_\tau$, the product of the conserved self-conjugacy norm $\kappa = \sqrt{5}$ (the invariant $\varphi + \varphi^{-1} = \sqrt{5}$ of the Majorana pair, the same κ that scales the holonomy amplitude feeding (34)) and the channel's native rate $\Gamma_\tau = \varphi/\sqrt{5}$: the pair-product rate $\omega_\xi \omega_\eta$ is referred to the single self-conjugate channel per unit of its own breathing rate. Both factors are fixed upstream; no new normalisation enters:

$$f_{\text{breath}} = \frac{\omega_\xi \cdot \omega_\eta}{\varphi \cdot \mathcal{S}^6} = \frac{\varphi^2/5}{\varphi \cdot 10^6} = \frac{\varphi}{5 \times 10^6} = (1 + \sqrt{5}) \times 10^{-7}. \quad (\text{G.59})$$

No free parameter enters: ω_ξ , ω_η , Γ_τ , $\mathcal{S}$, k_{obs} , n_{min} , and φ are all fixed by the rank-5 projection structure.

Independent structural verification. The global PLL loop gain at the critical node ($n = 5, k = 2$) is:

$$f_{\text{gain}} = \frac{C_{\text{bulk}}^{3/2}}{H_{\text{bdy}}} = \frac{(2\sqrt{5})^{3/2}}{3\pi} \approx 1.003463, \quad (\text{G.60})$$

where $C_{\text{bulk}} = 2\sqrt{5}$ is the bulk spectral curvature ($(\gamma^6)^2 = 5$, Section 3) and $H_{\text{bdy}} = 3\pi$ is the minimal Majorana holonomy on $S^2 \times S^1(\tau)$. The system is marginally above the synchronisation threshold $f_{\text{gain}} > 1$; the excess $\delta F = 1 - f_{\text{gain}}^{-1} \approx 0.345\%$ is the irreducible projection residue. This loop gain is a separate structural derivation; it provides an independent consistency check that the PLL is locked and that the back-action scale is indeed $O(10^{-7})$.

Remark 8 (The gain excess is the arrow of time). The loop-gain excess and the arrow-of-time residue are the same number: $\delta F \equiv 1 - f_{\text{gain}}^{-1} = 1 - 3\pi/(2\sqrt{5})^{3/2} = \delta_F \approx 3.451 \times 10^{-3}$, the fractional unitarity loss per projection cycle of Appendix H. By Lindemann–Weierstrass $\pi \notin \overline{\mathbb{Q}}(\sqrt{5})$, so $f_{\text{gain}} = 1$ exactly is impossible: the lock is necessarily marginal and $\delta_F \neq 0$ is exact, not approximate, the same transcendence that forbids $\varepsilon_7 = 0$ keeps the breathing from closing. The verification that the PLL sits above threshold is therefore not an auxiliary inequality; it is the irreversibility statement itself, read in the gain register.

The algebraic direction of f_{breath} is fixed by the exact identity in $\mathbb{Z}[\sqrt{5}]$:

$$\frac{2\varphi}{5 + 3\sqrt{5}} = \frac{5 - \sqrt{5}}{10} \in \mathbb{Z}[\sqrt{5}],$$

which shows $f_{\text{breath}} = (1 + \sqrt{5}) \times 10^{-7} \in \mathbb{Z}[\sqrt{5}] \cdot 10^{-7}$. The absolute scale 10^{-7} follows from eq. (G.59) without free parameters. $\square$

H Standard Model fields as projections of the Majorana bulk

This appendix shows how the field content of the Standard Model arises as the low-energy decomposition of the projected boundary connection and the projected Majorana spinor. The gauge-field skeleton, tree-level electroweak readout, and the one-step structural electroweak boundary matching are derived; the full fermionic transporter beyond the charged-lepton sector and the full scheme-dependent electroweak renormalisation-group evolution $\sin^2 \hat{\theta}_W(Q^2)$ remain open.

The four interactions from one projection. All four interaction sectors are represented as structural readouts of the single boundary equation $(\gamma^\mu D_\mu + \alpha \xi + i\alpha \eta \gamma^5) \Psi = \delta \mathcal{S}_{\ker} / \delta \bar{\Psi}$, with no free parameters in the gauge structure.

Interaction	Projection mechanism	Key equation	Status	Open
GR	Chiral limit $\eta \rightarrow 0$	$G_{\mu\nu} = \frac{1}{\alpha} \partial_\mu \Xi^\dagger \partial_\nu \Xi$	skeleton	R_5 , dynamics open
QED	Chiral limit $\xi \rightarrow 0$, $U(1) \rightarrow U(1)_{\text{disc}}$	$\partial_\nu F^{\mu\nu} = J^\mu$, step π/n	skeleton	running $\alpha(Q^2)$
EW	$\xi \sim \eta$, rotation B_μ / W_μ^3	$\sin^2 \theta_W = 3/13$	derived	$\sin^2 \theta_W(Q^2)$, masses
QCD	$n_{\text{conf}} = 7 \in \ker \Pi$, $K_7 \cong \mathbb{C}^3$	$\mathfrak{su}(3)_c$ from pairwise $SU(2)$	structural	full Yang–Mills, 3 gen.

GR and QED are the two chiral limits of the same field equation at the boundary: GR from the longitudinal sector (ξ , massive, continuous $SU(2)$) and QED from the transverse sector (η , radiative, discretised $U(1)$). The electroweak interaction lives at the intermediate scale $\xi \sim \eta$ where both channels are active simultaneously; the mixing angle $\sin^2 \theta_W = 3/13$ is an unconditional theorem of $\text{TL}_4(\varphi)$ (Section 6), not a fitted parameter. QCD arises from the confined mode $n = 7$: the unique element of $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$ that does not factorise within the Fibonacci basis $\{2, 3, 5\}$ and is therefore forced into $\ker \Pi$, where its triplet structure $K_7 \cong \mathbb{C}^3$ closes to $\mathfrak{su}(3)_c$. Confinement is not a dynamical consequence but a projective necessity: $[5]_q = 0$ in $\text{TL}_4(\varphi)$ truncates the $n = 7$ sector exactly at the boundary.

Coupling constants: what is derivable. The four coupling constants of the Standard Model have distinct statuses within the present construction. In the SM all four are measured inputs with no theoretical derivation. In the present construction they are all structural outputs of the projector Π , but with different degrees of completeness.

Constant	Structural derivation	Value	Status	Open
α^{-1}	Pole of $H_\alpha(z)$, zero parameters	137.035999170	derived	,
$\sin^2 \theta_W$	Theorem of $\text{TL}_4(\varphi)$: $3/(2r+3)$; closed by ξ -channel eq. (43)	0.23125	closed	RG evolution $\sin^2 \hat{\theta}_W(Q^2)$
$\alpha_s(M_Z)$	$SU(3)$ structure derived; value not yet	—	structural	Yang–Mills, running
G_N	$G = R_5^2 c^3 (\pi \alpha)^2 / \hbar$; R_5 not derived	consistency 0.14%	structural	R_5 from first principles

The key structural distinction is the following. α^{-1} is the most complete derivation because it is the pole of the filter itself, the stable operating point that the boundary *cannot not have* given the rank-5 projection. $\sin^2 \theta_W = 3/13$ is the unconditional tree-level boundary value; the one-step ξ -channel

structural closing gives $\sin^2 \theta_W = 0.23125$, within 0.75σ of the Z-pole value. The one-step boundary electroweak matching is closed at $\sin^2 \theta_W = 0.23125$. The electroweak scale v_{EW} and Fermi constant G_F are derived in the companion Heegner–Yukawa paper [9], conditional on R_∞ . The full scheme-dependent $\overline{MS}$ running function $\sin^2 \hat{\theta}_W(Q^2)$ — including electroweak beta functions, loop thresholds, and scheme conversion — is not derived here. α_s and G_N require additional structural steps (the Yang–Mills action for the confined sector and the derivation of R_5 from α and φ alone) that remain open. The four coupling sectors are organised by the same projector Π , with different degrees of closure: α^{-1} and $\sin^2 \theta_W$ are derived; α_s and G_N require further structural steps. The same object $\mathbf{M}^2$ that fixes α^{-1} also fixes $\sin^2 \theta_W$: the two are two scalar contractions of one Fibonacci matrix.

H.1 Boundary covariant derivative and gauge fields

The boundary field $\Psi = \Pi \Xi$ couples to a covariant derivative

$$D_\mu^{\text{bdy}} = \partial_\mu + \mathcal{A}_\mu,$$

where the boundary connection decomposes along the three boundary sectors:

$$\mathcal{A}_\mu = A_\mu Q + W_\mu^a T_a + G_\mu^A T_A.$$

Here A_μ is the radiative phase connection of the η -channel ($U(1)$ EM), W_μ^a ($a = 1, 2, 3$) are the chiral connections of the ξ -channel ($SU(2)$), and G_μ^A ($A = 1, \dots, 8$) are the confined connections of the kernel sector ($SU(3)_c$, from the $n = 7$ triplet $K_7 \subset \ker \Pi$, Section 6.3).

The gauge field strengths arise from the curvature of D_μ^{bdy} :

$$\mathcal{F}_{\mu\nu} = [D_\mu^{\text{bdy}}, D_\nu^{\text{bdy}}] = F_{\mu\nu} Q + W_{\mu\nu}^a T_a + G_{\mu\nu}^A T_A.$$

These are not postulated independently. They are the three curvature components of a single projected connection.

H.2 Fermionic content

The observable fermions are chiral projections of the boundary field:

$$\psi_f = P_f P_{L/R} \Pi \Xi, \quad P_{L/R} = \frac{1}{2}(1 \mp \gamma^5),$$

where P_f projects onto the appropriate boundary sector. The decomposition $\Psi = \Psi_{\text{lept}} \oplus \Psi_{\text{q}} \oplus \Psi_{\text{conf}}$ identifies: leptons as modes crossing the boundary without confined charge, quarks as modes coupled to the $n = 7$ kernel sector, and Ψ_{conf} as the confined component that does not produce a free boundary state. No fermion is postulated independently; every fermion field is a chiral-projected component of the Majorana bulk Ξ .

H.3 Effective scalar field

The Higgs field is not a fundamental input. It is the local order parameter of the boundary lock-in ξ/η :

$$\Phi_H(x) \sim (\rho(x), u(x)), \quad \rho^2(x) \sim \xi(x),$$

where ρ is the coherent modulus and u is the chiral orientation. The effective potential is

$$V_{\text{eff}}(\Phi_H) = V_{\text{lock}}(\xi, \eta) + V_{\text{ker}}(\Psi), \quad V'_{\text{ker}}(|\Psi|) = \frac{\delta \mathcal{S}_{\text{ker}}}{\delta |\Psi|} \propto |\Psi|^{2/3}.$$

The full electroweak potential and its symmetry-breaking pattern are not yet derived; they constitute an open problem in future work.

H.4 Electroweak mixing

The tree-level electroweak mixing is a rotation between the η -channel phase connection B_μ and the neutral ξ -channel connection W_μ^3 :

$$\begin{pmatrix} A_\mu^{\text{em}} \\ Z_\mu \end{pmatrix} = \begin{pmatrix} \cos \theta_W & \sin \theta_W \\ -\sin \theta_W & \cos \theta_W \end{pmatrix} \begin{pmatrix} B_\mu \\ W_\mu^3 \end{pmatrix}, \quad \sin^2 \theta_W = \frac{3}{13}.$$

The charged bosons are $W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \mp iW_\mu^2)$. The value $\sin^2 \theta_W = 3/13$ is derived unconditionally from the rank-5 projection (Section 6), not fitted.

H.5 Thermodynamic correspondence

The boundary projection admits a natural thermodynamic reading of the boundary dynamics. The kernel entropy functional $\mathcal{S}_{\text{ker}}[\Psi] = -\text{Tr}_{\mathcal{K}_{\text{ret}}^{(5)}}(\rho_\Psi \log \rho_\Psi)$ plays the role of the *free energy* of the projected boundary: the nonlinear return term $\delta \mathcal{S}_{\text{ker}} / \delta \bar{\Psi}$ is its functional gradient, driving the boundary toward the stable operating point Ψ_* . The breathing recursion $B_{n+1} = 2B_n^{2/3}$ is the discrete RG flow toward the attractor $B_* = 8$, with exact beta-function $\beta(u) = -\frac{1}{3}(u - \log 8)$ and Lyapunov exponent $\lambda = 2/3$. The fractional unitarity loss per projection cycle, $\delta_F = 1 - 3\pi/(2\sqrt{5})^{3/2} \approx 3.45 \times 10^{-3}$, plays the role of the irreversible entropy production rate: it is non-zero because $\pi \notin \overline{\mathbb{Q}}(\sqrt{5})$ (Lindemann–Weierstrass), making the boundary dynamics permanently irreversible. The arrow of time is the monotone increase of the logarithmic phase $\theta_{\log} = \frac{1}{2} \log(\eta/\xi)$ at rate $\dot{\theta}_{\log} = \Gamma_\tau$.

SM field	Projection origin	Status
A_μ (photon)	η -channel connection, $n = 5$ zero mode	structural
$W_\mu^\pm, Z_\mu$	ξ -channel + EW mixing $\sin^2 \theta_W = 3/13$	tree-level
G_μ^A (gluons)	$n = 7$ kernel triplet $K_7 \subset \ker \Pi$	structural
ψ_f (fermions)	chiral projections $P_f P_{L/R} \Pi \Xi$	structural
Φ_H (Higgs)	local order parameter of lock-in ξ/η	structural
A_s (CMB amplitude)	Heegner CM transfer at τ_{163} [8]	structural
charged-lepton Yukawa	Φ_Y^ℓ closed structurally; absolute masses conditional on R_∞ ; full fermion sector (quarks, CKM, neutrinos) open	conditional / see [9]
EW running	full scheme-dependent $\overline{\text{MS}}$ RG evolution $\sin^2 \hat{\theta}_W(Q^2)$ beyond the one-step boundary matching	Open
QCD confinement	$n = 7 \in \ker \Pi$, $[5]_q = 0$ truncation	structural
Arrow of time	$\dot{\theta}_{\log} = \Gamma_\tau > 0$, $\delta_F \neq 0$	structural
Entropy production	$\mathcal{S}_{\text{ker}}$, $\lambda = 2/3$ Lyapunov	structural